

# Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 270 factors, ordered by in-sample L-S CAGR (last 48 months, net of cost).

## HOW TO READ

■ Long ■ Short ■ Neutral

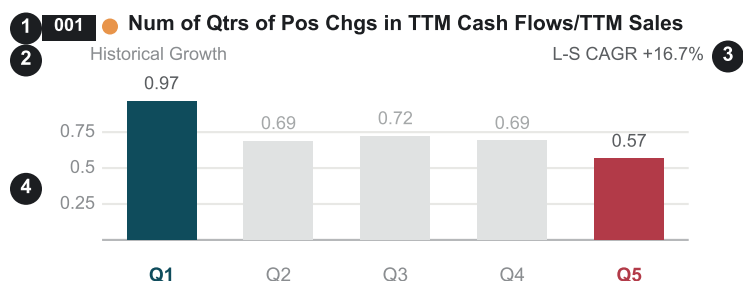
Quintiles are selected as long or short by performance against a spread-based threshold — not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

## FACTOR STYLES

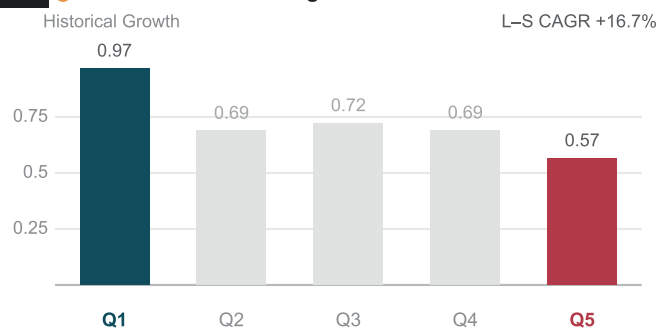
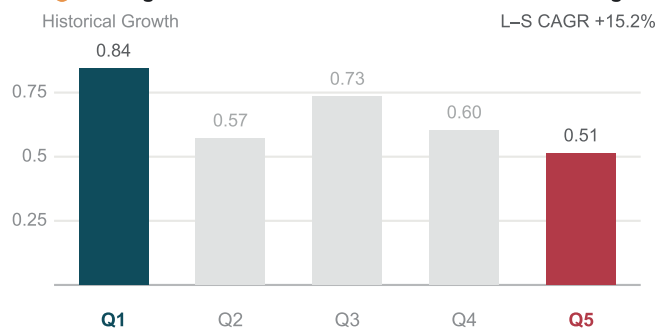
|                      |    |                   |    |
|----------------------|----|-------------------|----|
| Valuation            | 41 | Price Momentum    | 56 |
| Earnings Quality     | 55 | Size              | 1  |
| Analyst Expectations | 24 | Historical Growth | 55 |
| Capital Efficiency   | 28 | Volatility        | 10 |

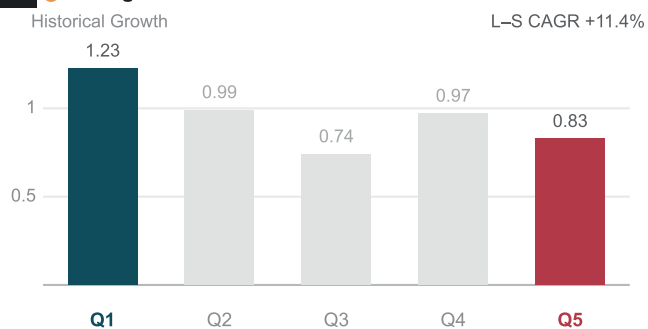
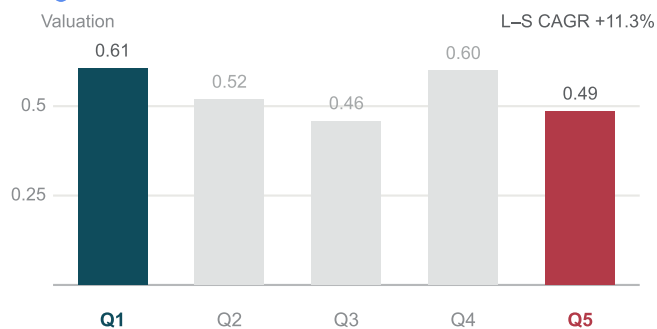
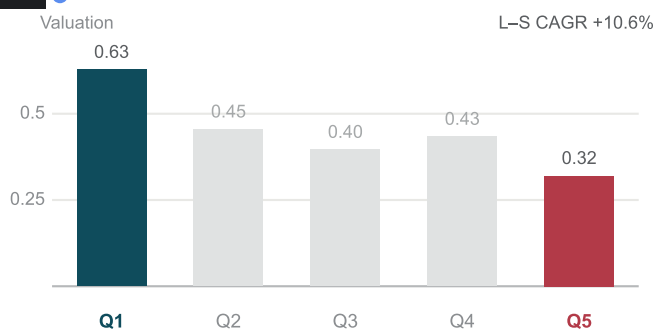
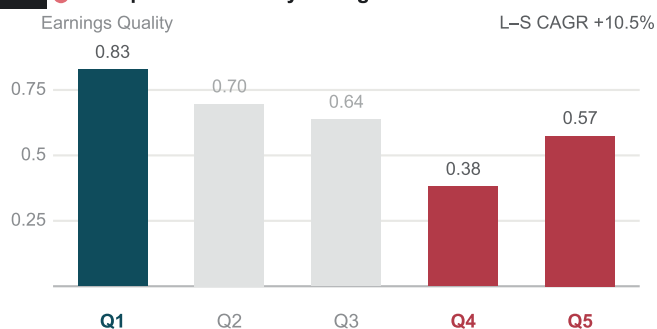
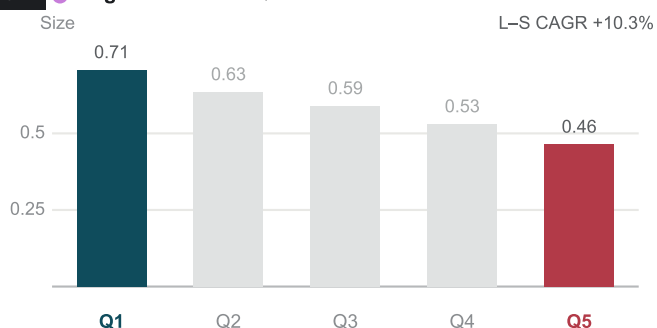
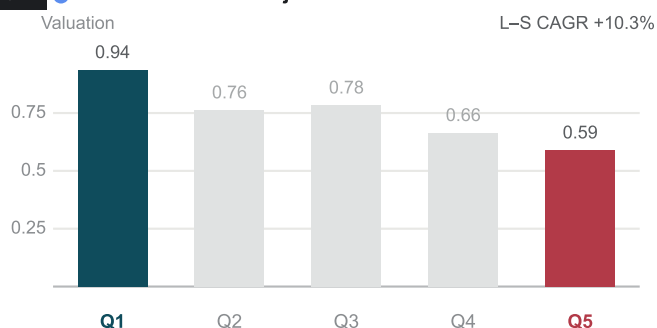
**001** In current model portfolio (50 factors)

## READING GUIDE — FIRST FACTOR AS EXAMPLE

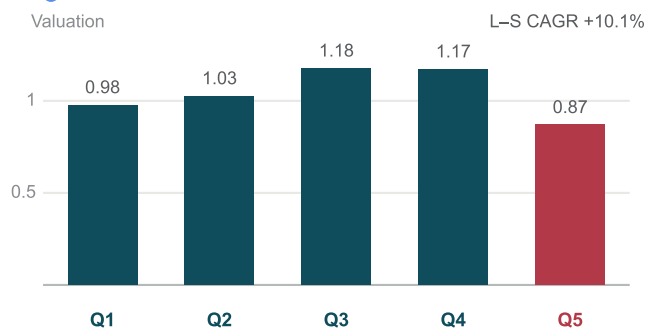


- 1** Rank - ordered by (3) L-S CAGR, descending.  
Filled dark badge = in the current Top-50 portfolio.
- 2** Factor name - colour dot = style (legend above).
- 3** L-S CAGR - LAST 48 MONTHS only (rolling IS window), net of 10 bp cost. Selection / sort basis - NOT the chart period.
- 4** Bars - FULL HISTORY (Jul 2015 - Jul 2026): avg monthly return of the five quintiles, BEFORE costs.  
So bars can look strong while (3) is negative (weak recent signal / high turnover), and vice versa.

**001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales****002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Pretax Income Per Share to Price****004 ● Current Liability to Price****005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flo****006 ● Operating Earnings to Assets Ratio**

**007 ● Change of Asset Turnover****008 ● Assets to Price Ratio****009 ● Sales to Price Ratio****010 ● Unexpected Inventory Change****011 ● Log of Total Last Quarter Assets****012 ● Inverse P/E Ratio Adj for Growth and Yield**

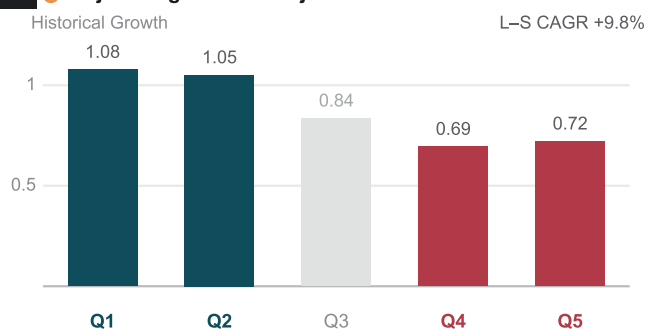
## 013 ● Indicated Annual Dividends to Price Ratio



## 014 ● Cash to Price



## 015 ● Adj 1Y Chg in Asset Adjusted FCF



## 016 ● Long Term Debt to Equity Ratio

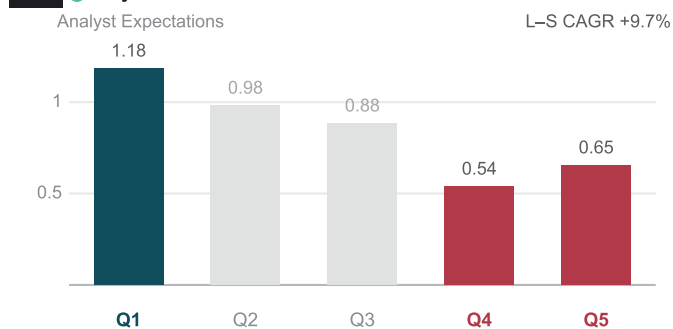
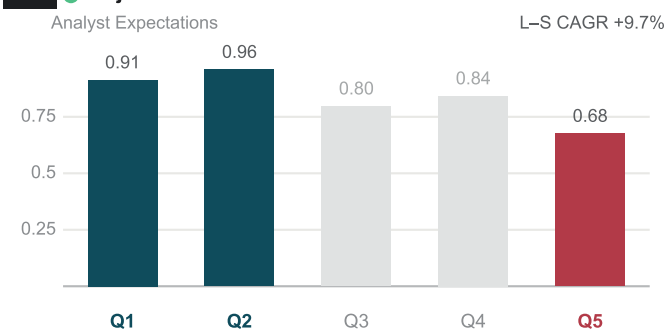
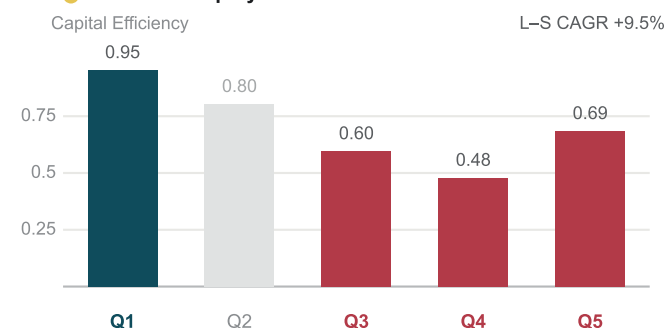
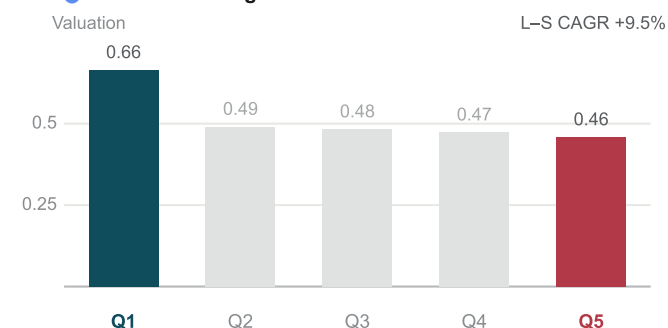


## 017 ● Return on Assets

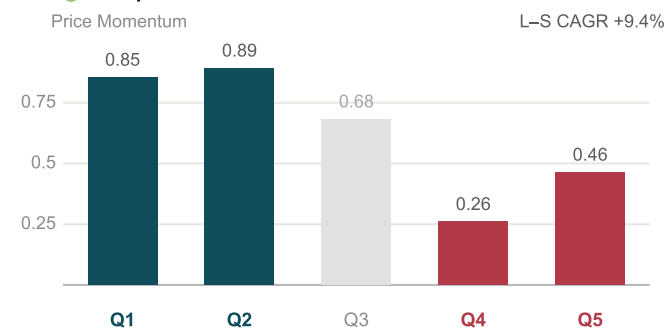


## 018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...



**019 ● Gross Profit Margin****020 ● Adjusted 3M Revision in FY1 EPS Estimate****021 ● Adjusted Inverse PEG****022 ● Analyst Dispersion for FY2 EPS****023 ● Return on Equity****024 ● Forward Earnings to Price**

## 025 ● Sharpe Ratio



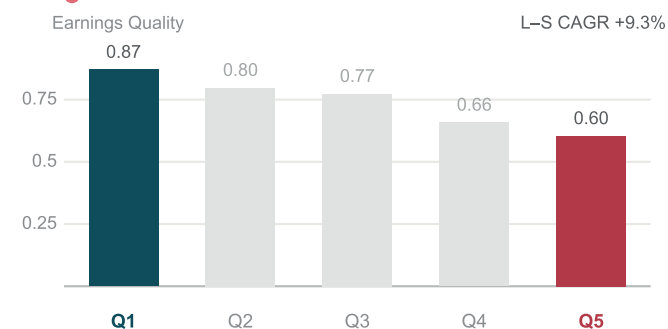
## 026 ● 12M Realized Price Volatility



## 027 ● Return on Invested Capital



## 028 ● Quick Ratio

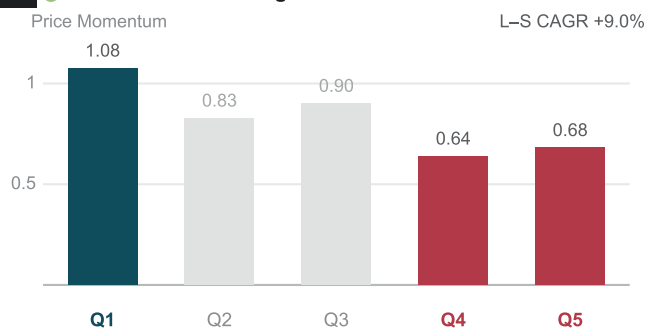
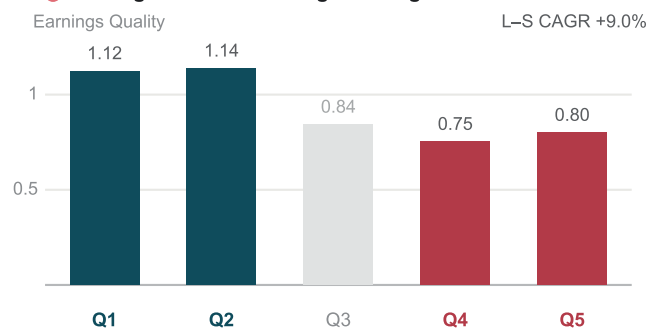
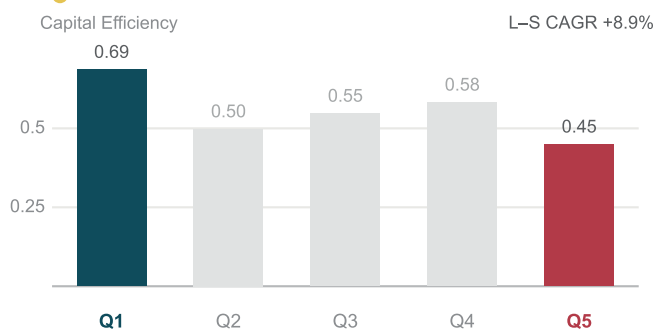
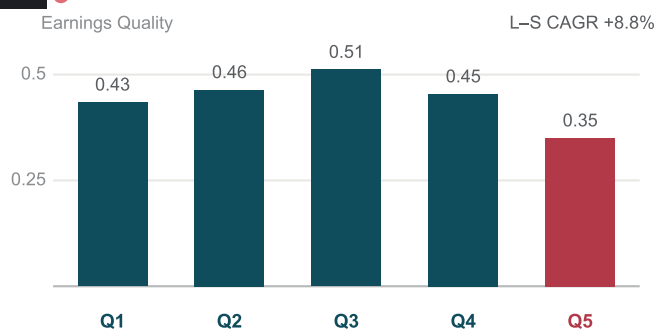
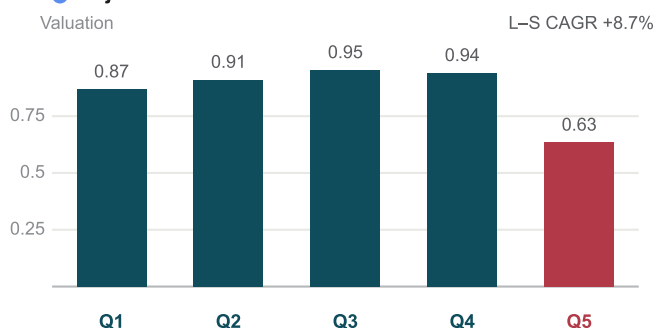
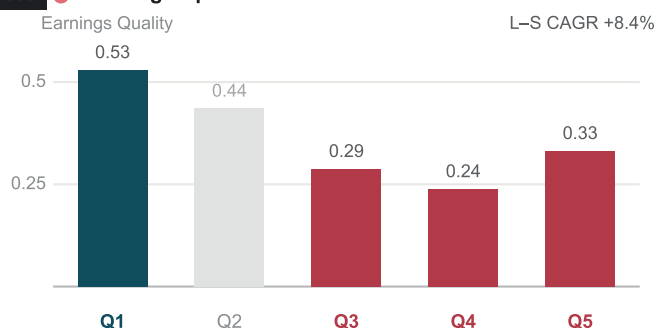


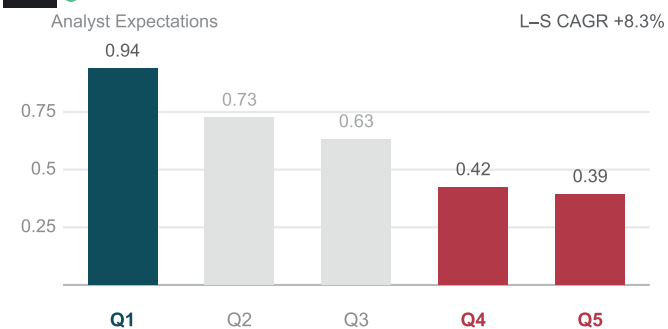
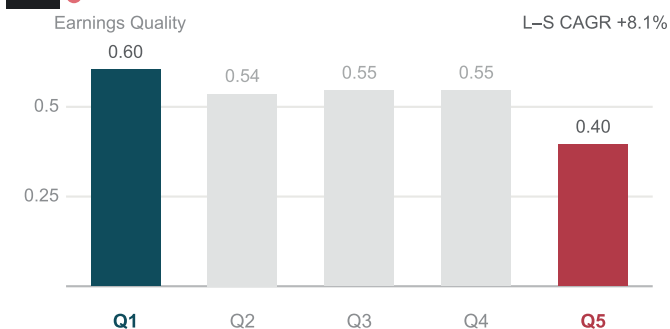
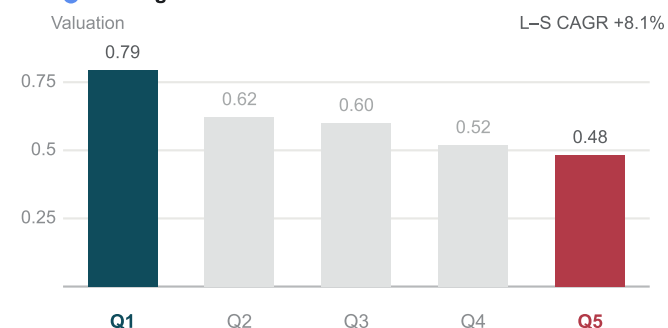
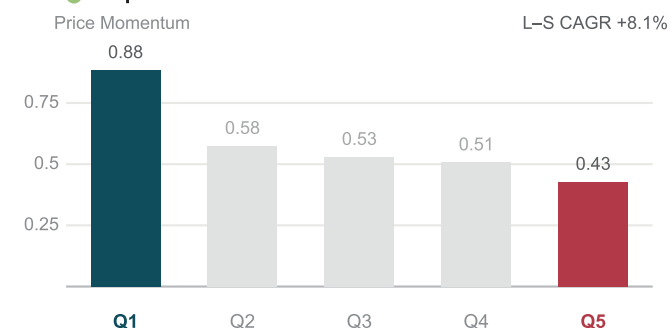
## 029 ● 3Y Chg in Operating Cash Flow to Price



## 030 ● Operating Earnings to Price Ratio

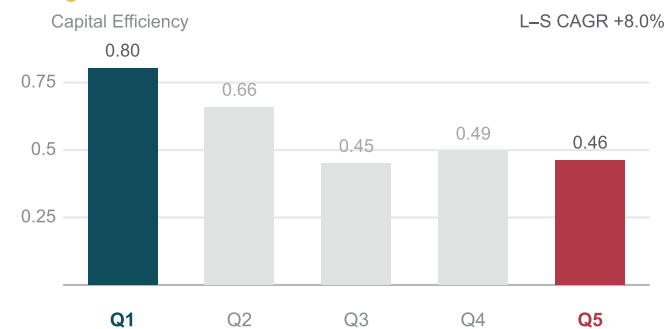


**031 ● Short Interest Trading Volume****032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales****033 ● Tobin's Q Ratio****034 ● ROA 20 Qtr Standard Deviation****035 ● Adj Growth Flow to Price****036 ● Working Capital to Sales**

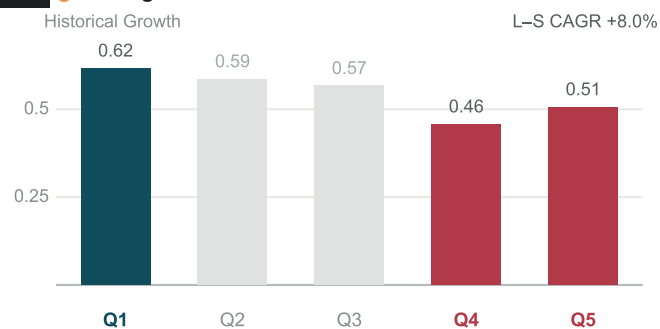
**037 ● 3M Revision in FY1 EPS Estimate****038 ● Adjusted Forward Earnings to Price****039 ● Normalized EP****040 ● Balance Sheet Accruals****041 ● Earnings to Price****042 ● Slope of 52W Price Trend**



## 043 ● EBITDA to Assets



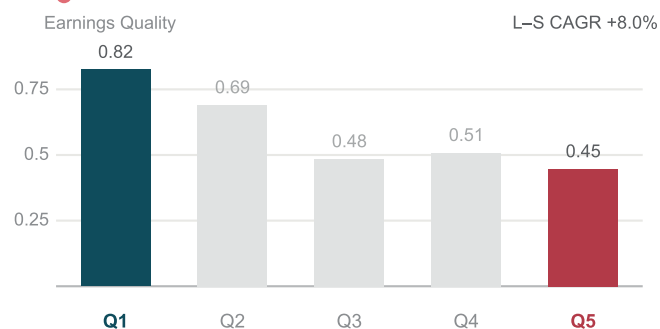
## 044 ● 3Y Chg in Cash Flow to Price



## 045 ● Consecutive Quarters of Declines in TTM Overhead/Sales



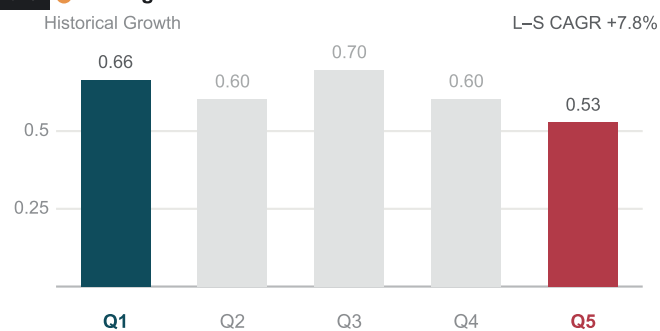
## 046 ● Cash Flow to Total Asset

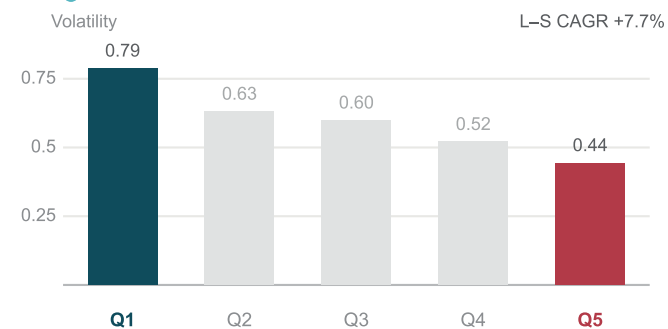
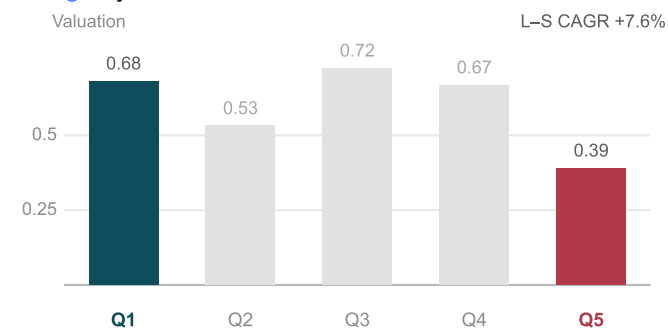
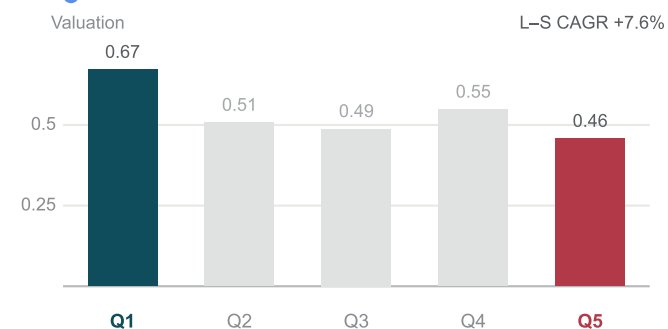
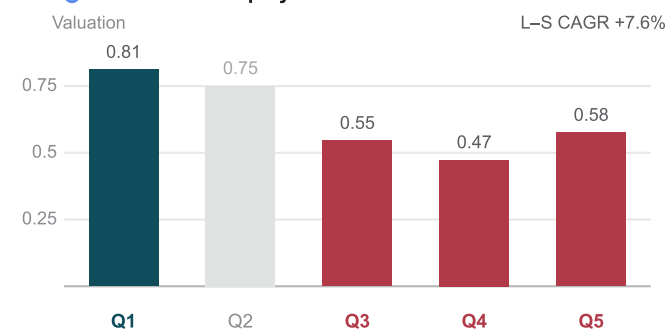
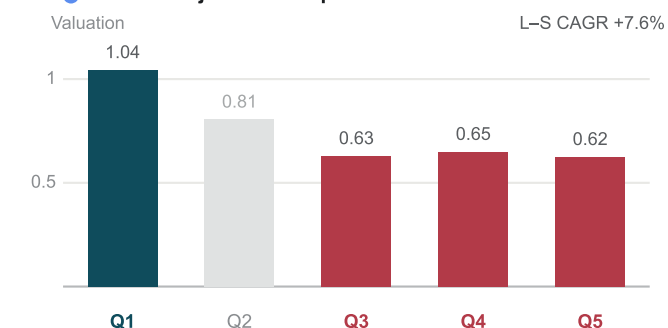


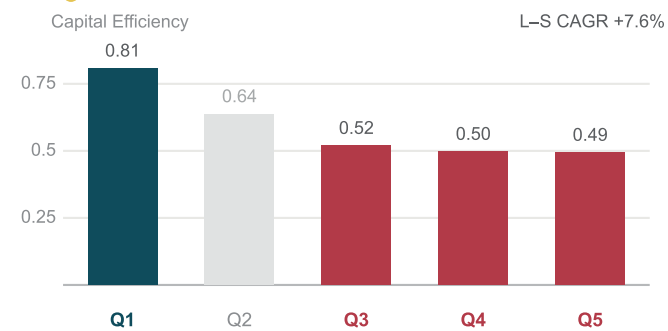
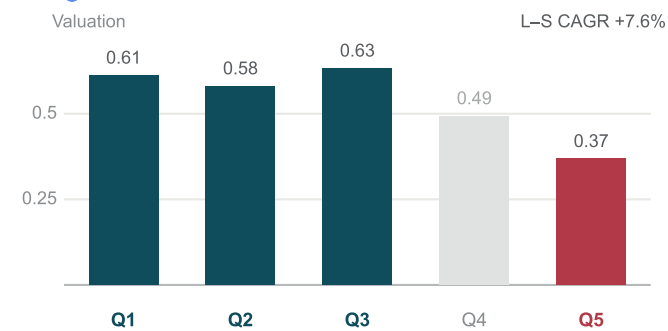
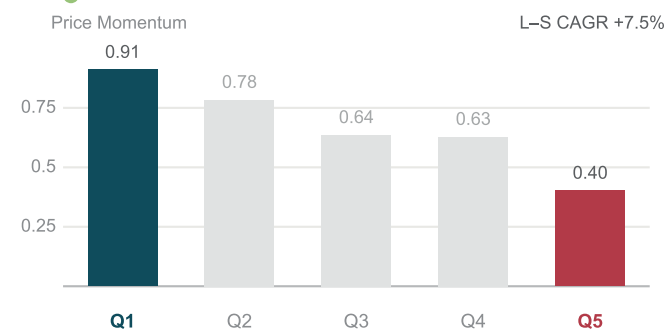
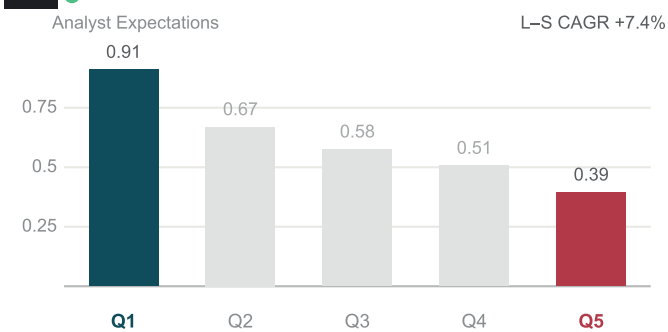
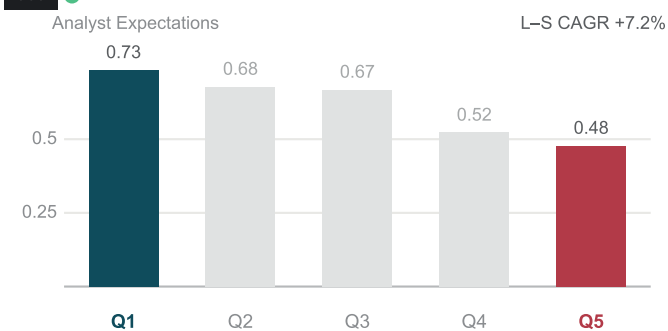
## 047 ● Book Leverage



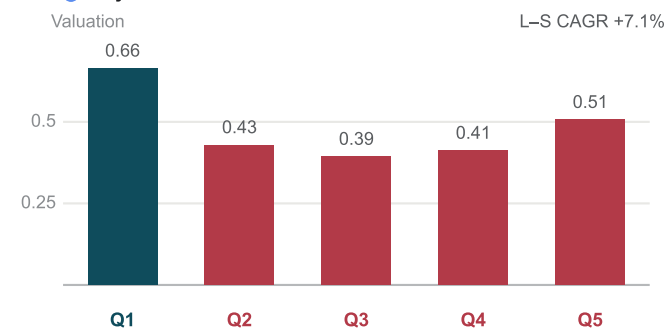
## 048 ● 1Y Chg in Sales to Price



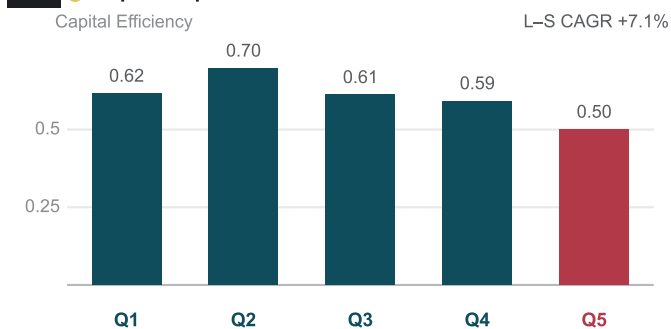
049 ● **60M CAPM Beta**050 ● **Adj Free Cash Flow to Price**051 ● **Sales to EV Ratio**052 ● **Cash Flow to Equity**053 ● **Cash to Adjusted Enterprise Value**054 ● **Expected LTG**

055 ● **EBIT to Assets**056 ● **Dividends to Price Ratio**057 ● **Short Interest Ratio**058 ● **3M Revision in FY2 EPS Estimate**059 ● **Amihud Illiquidity Measure**060 ● **Inverse PEG**

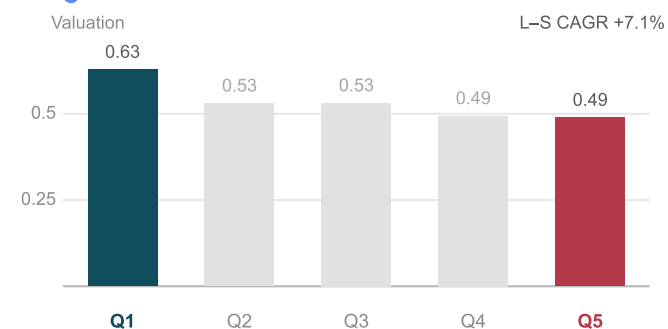
## 061 ● Adjusted EBIT to Price



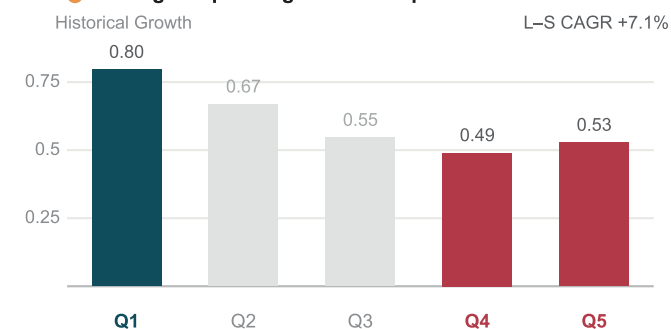
## 062 ● Capital Expenditure to Total Assets



## 063 ● Net Cash Flow to Price



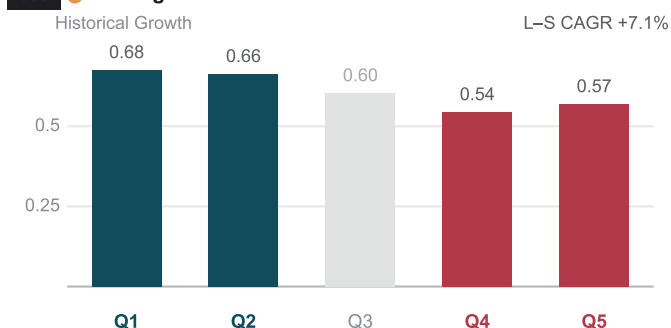
## 064 ● 3Y Chg in Operating Cash Flow per Share



## 065 ● 1Y Chg in Operating Margin



## 066 ● 1Y Chg in Cash Flow to Price



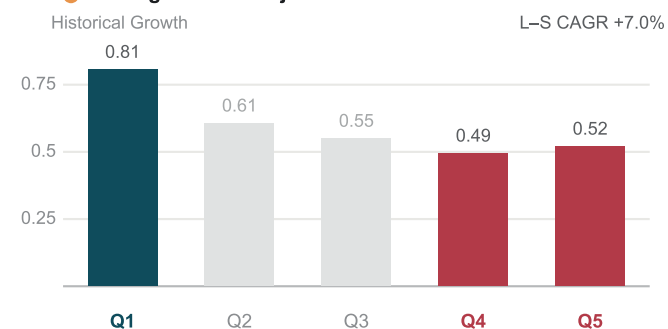
## 067 ● 50 Day to 200 Day Stock Price



## 068 ● Adjusted Book to Enterprise Value



## 069 ● 1Y Chg in Asset Adjusted EPS



## 070 ● Earnings Surprise

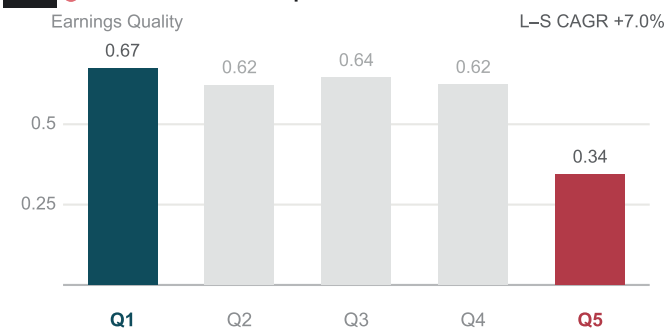
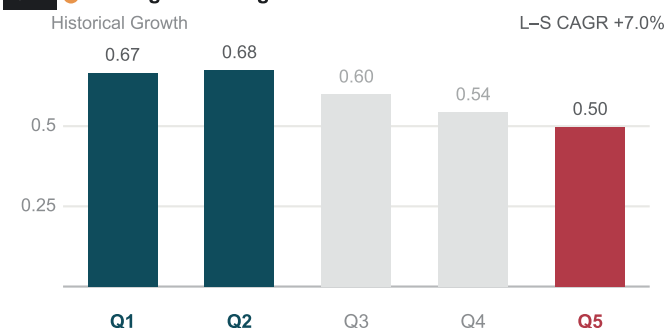


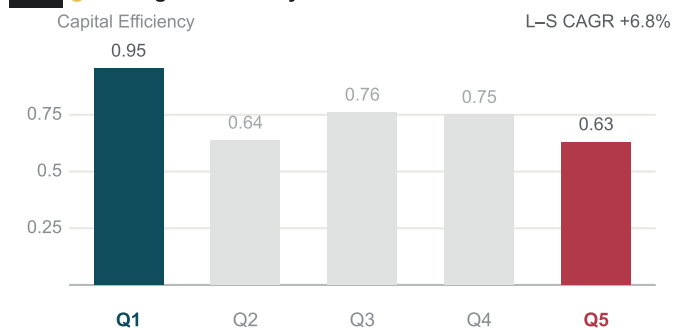
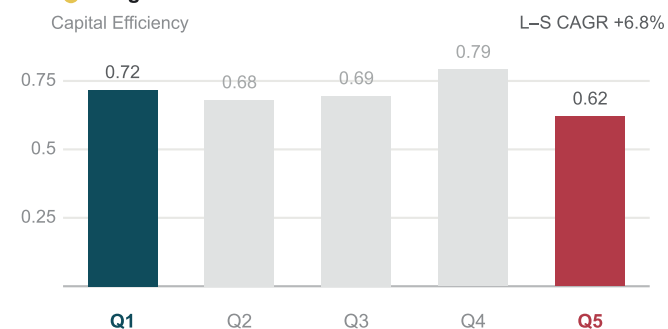
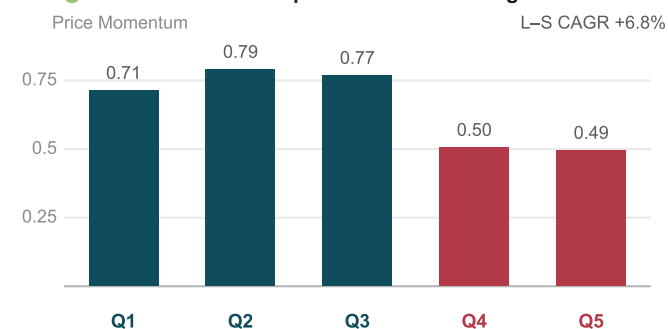
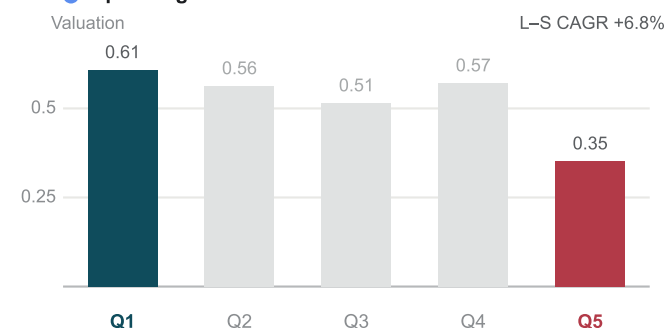
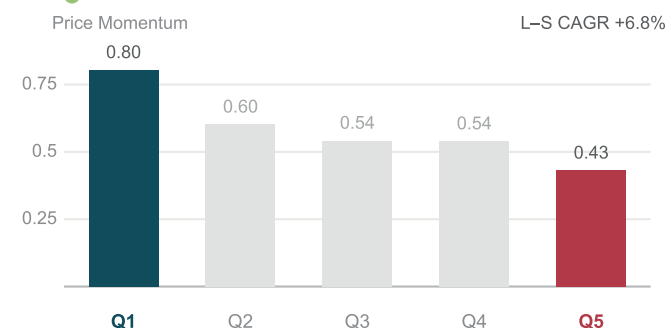
## 071 ● Street Revision Magnitude

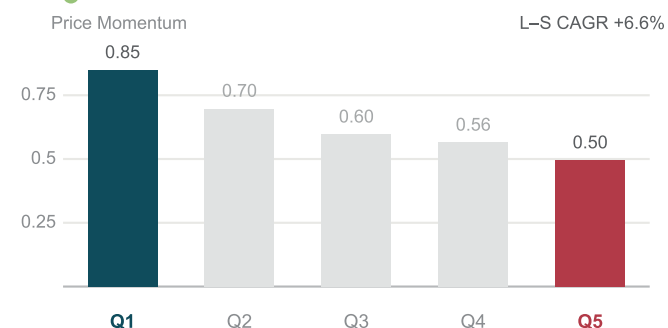
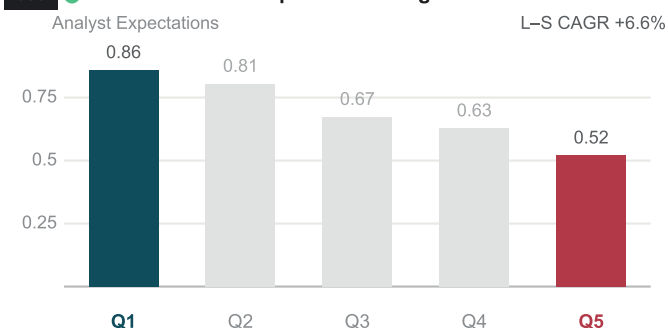


## 072 ● Cash Flow to Invested Capital

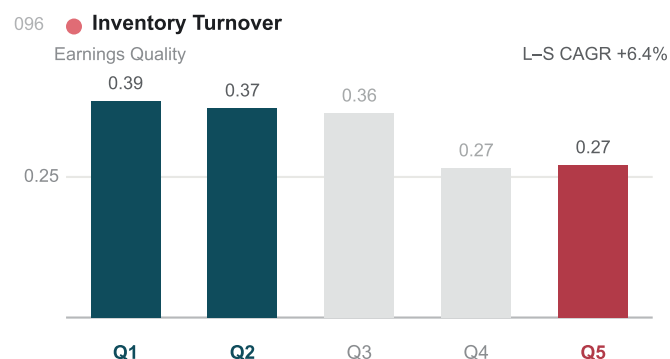
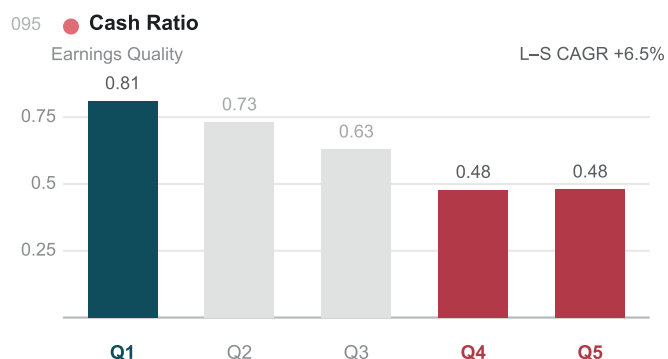
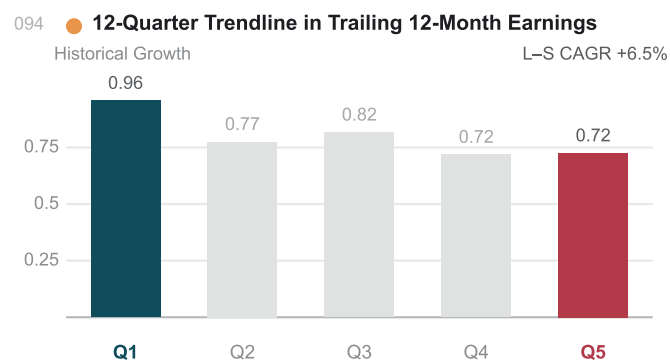
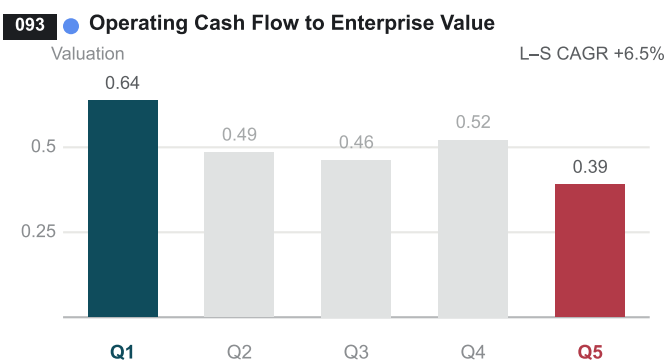
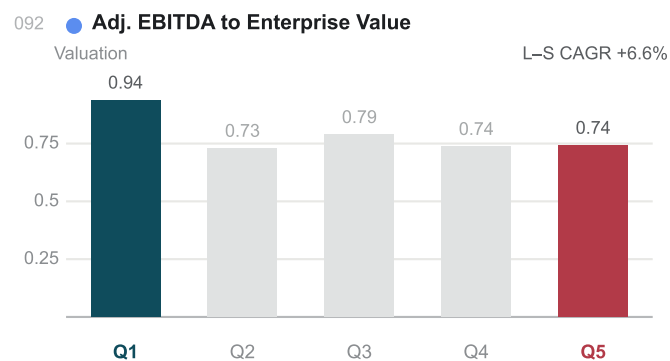
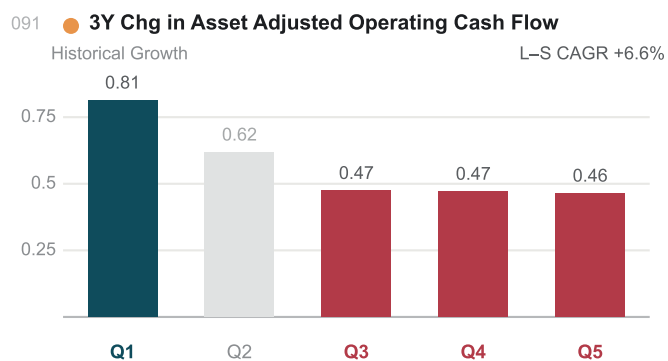


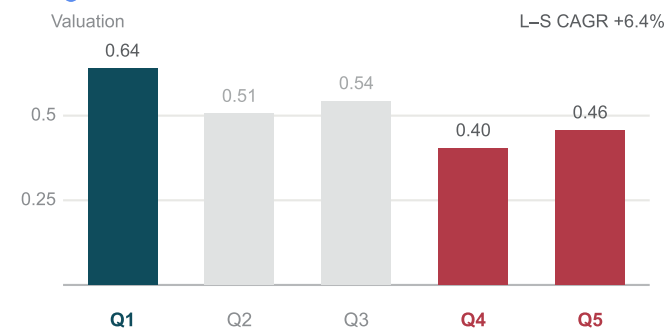
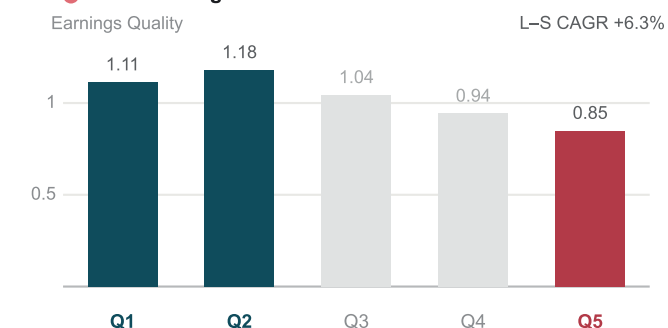
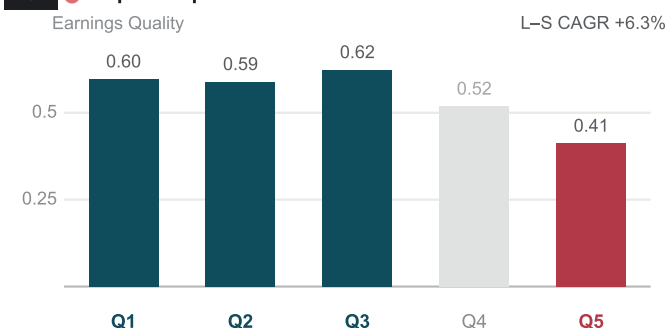
**073 ● Sales to Invested Capital****074 ● 39W Lagged Return****075 ● Operating Cash Flow to Invested Capital****076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow****077 ● 1Y Chg in Earnings to Price****078 ● Operating Cash Flow to Asset**

079 ● **EBITDA to Enterprise Value**080 ● **1Y Chg in Stock Buybacks**081 ● **Long Term Debt to Assets Ratio**082 ● **Institution Ownership Level - Active Manager**083 ● **Operating Cash Flow to Price**084 ● **15/36 Week Price Ratio**

085 ● **26W Relative Price Strength**086 ● **Adj 1Y Chg in Earnings per Share**087 ● **90 Day Coefficient of Variation**088 ● **Solvency Ratio**089 ● **9 Month Price Momentum**090 ● **Standardized Unexpected Earnings**





097 ● **Book to Price**098 ● **6M Chg in Target Price**099 ● **Adjusted Revision Magnitude**100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**101 ● **EBITDA Margin**102 ● **Capital Expenditure to Sales**

## 103 ● Free Cash Flow to Enterprise Value



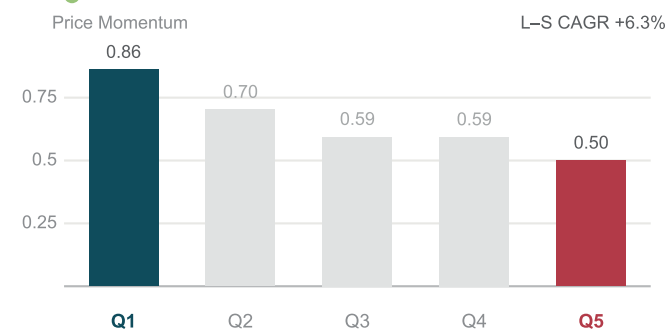
## 104 ● 1M Realized Price Volatility



## 105 ● EBITDA to Price



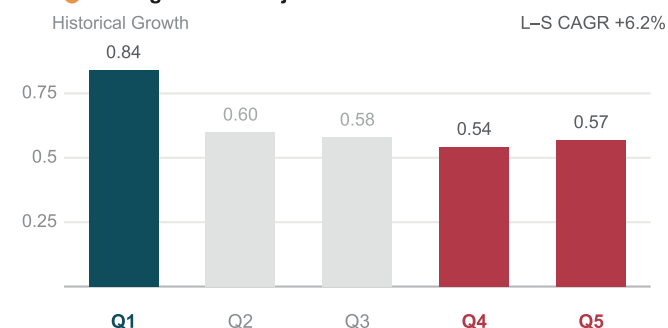
## 106 ● 4W to 52W Price Oscillator



## 107 ● Percent Above 260-Day Low



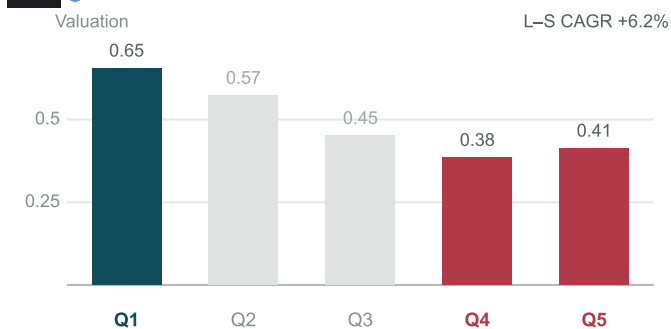
## 108 ● 1Y Chg in Asset Adjusted Cash Flow



## 109 ● Growth Flow to Price



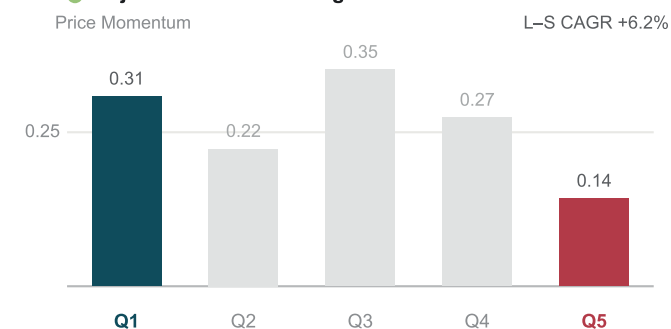
## 110 ● Cash to EV Ratio



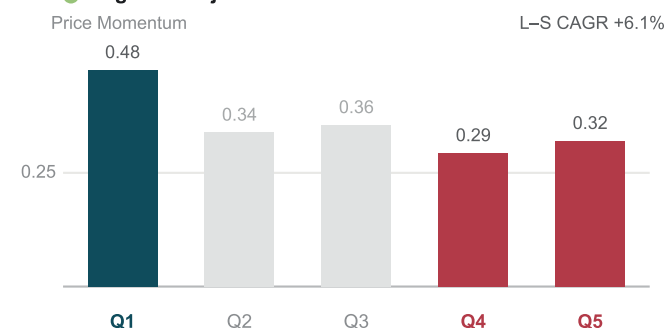
## 111 ● Year over Year Change in SGA to Sales



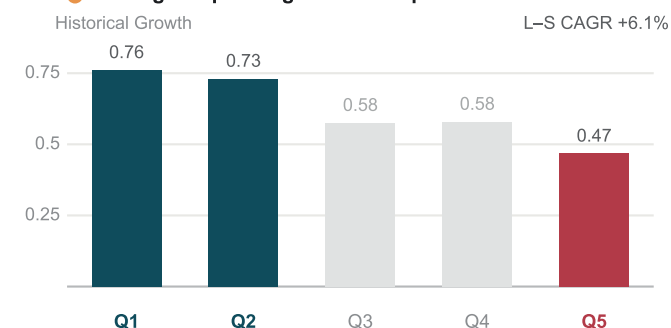
## 112 ● Adjusted 6 Month Average Share Turn Over

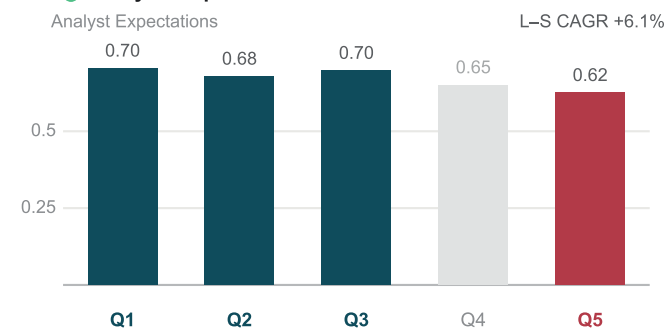
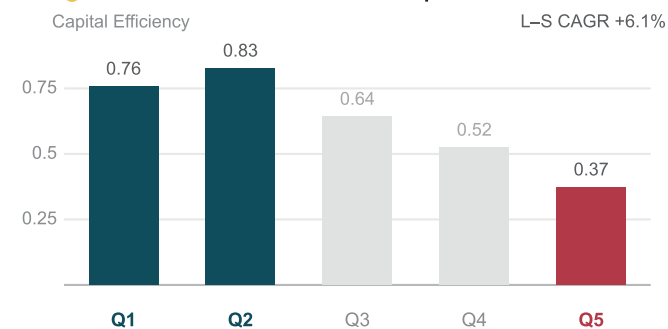
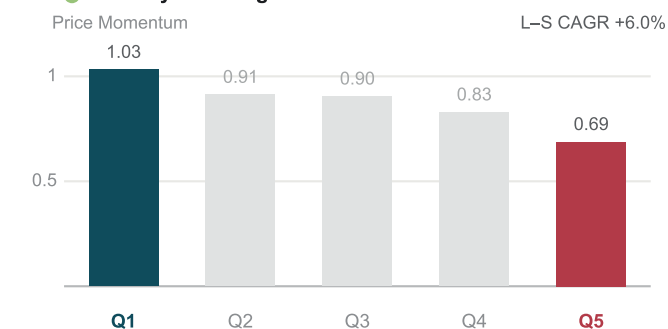


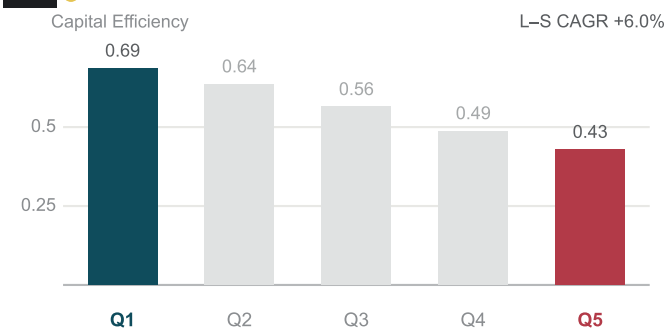
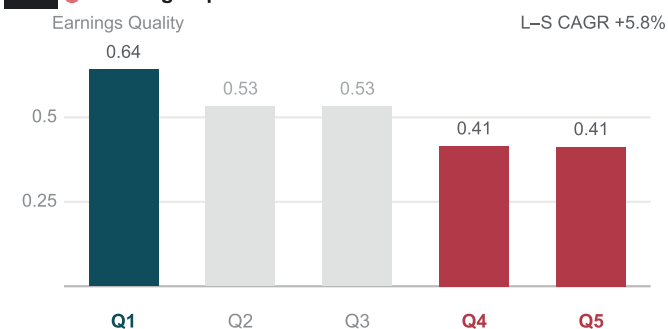
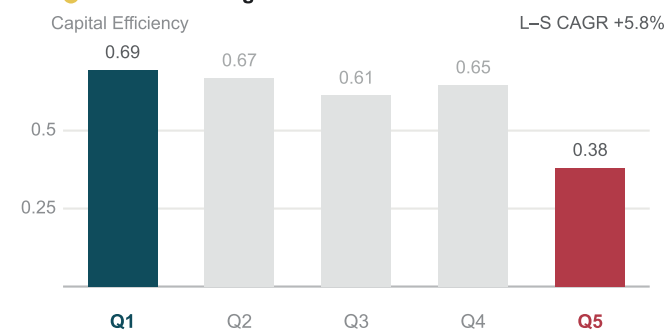
## 113 ● Log of Unadjusted Stock Price

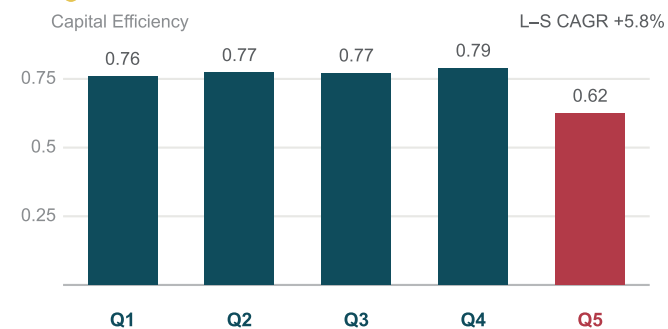
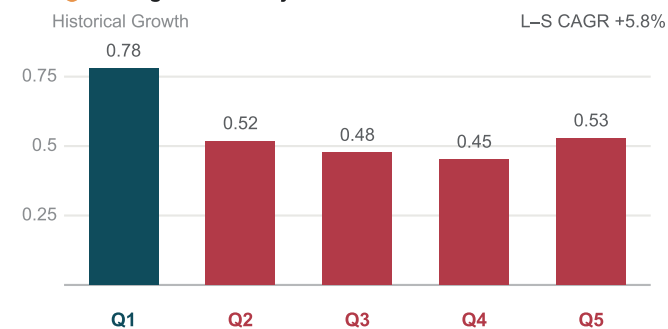
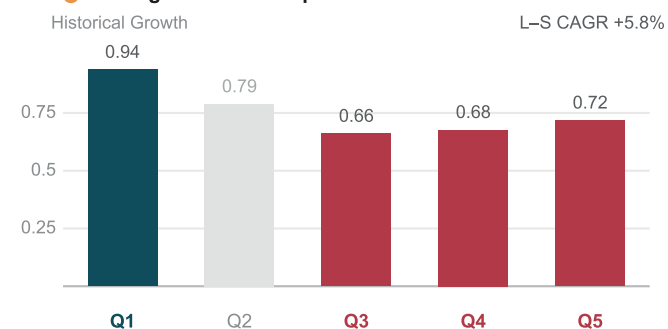
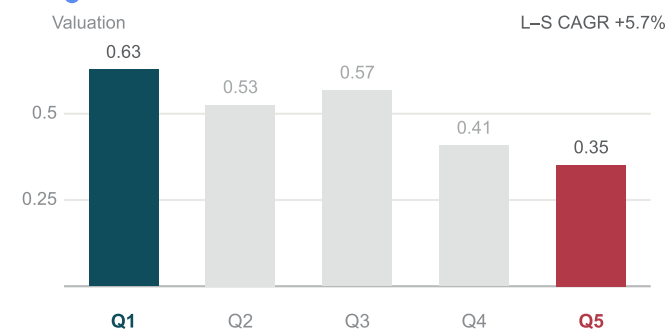
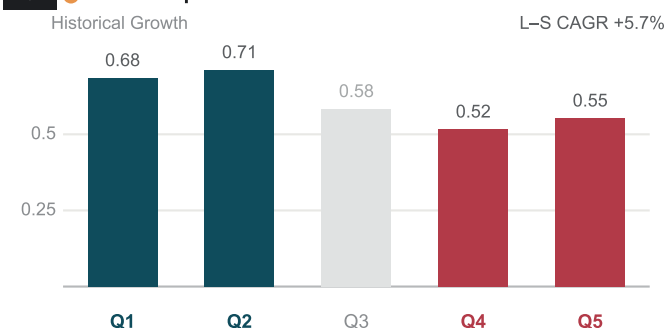


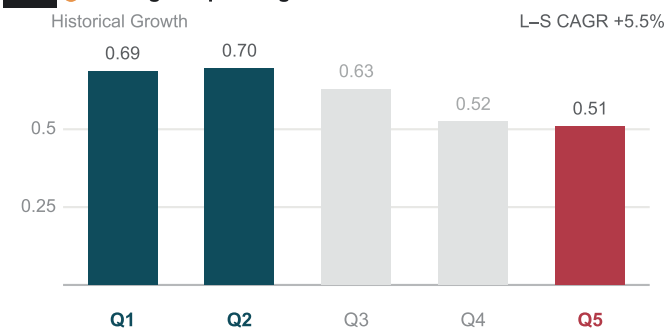
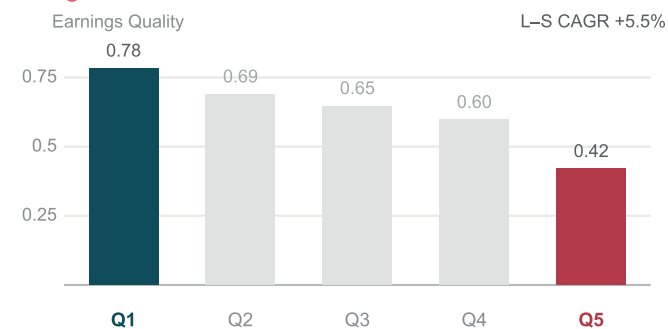
## 114 ● 1Y Chg in Operating Cash Flow per Share

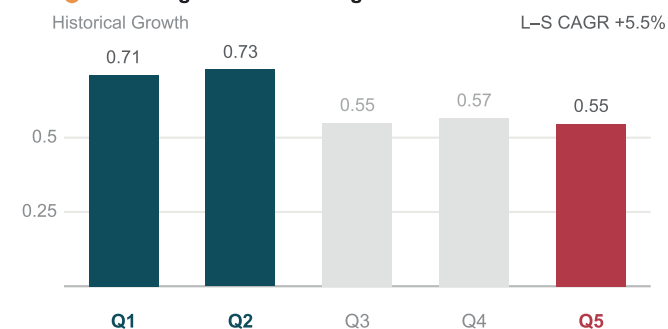
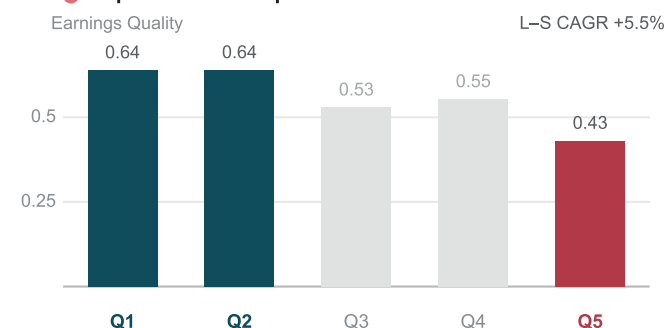
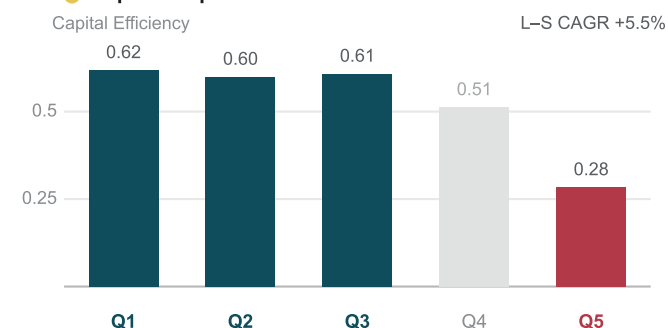


115 ● **Analyst Dispersion for FY1 EPS**116 ● **Cash Flow Return on Invested Capital**117 ● **1Y Chg in Asset Adjusted Free Cash Flow**118 ● **Stability of Change in Number of Holders**119 ● **Operating Cash Flow Ratio**120 ● **Equity Turnover**

**121 ● Receivables Turnover Ratio****122 ● Adjusted Interest Coverage Ratio****123 ● 3Y Chg in Free Cash Flow to Price****124 ● Working Capital Accruals****125 ● Interest Coverage Ratio****126 ● Forward Free Cash Flow to Price Ratio**

127 ● **Debt to Assets Ratio**128 ● **3Y Chg in Asset Adjusted Cash Flow**129 ● **3Y Chg in Cash Flow per Share**130 ● **Free Cash Flow to Price**131 ● **Dividend per Share Growth**132 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

**133 ● 1Y Chg in Operating Cash Flow to Price**

**134 ● Cash Burn Rate**

**135 ● 52-Week Volume Price Trend with 20-Day Lag**

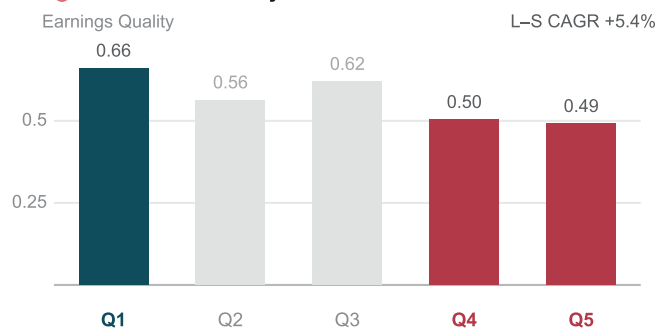
**136 ● 3Y Average Annual Earnings Growth**

**137 ● Depreciation to Capex Ratio**

**138 ● Capital Acquisition Ratio**




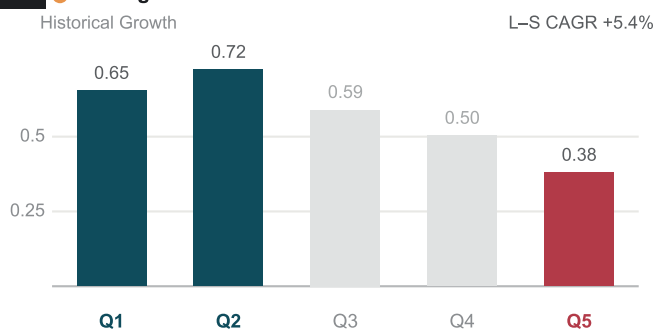
## 139 ● Free Cash Flow Stability



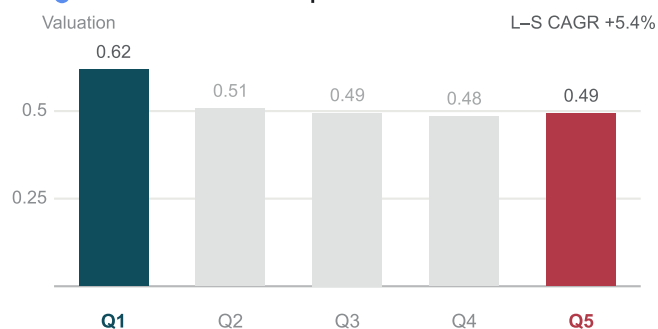
## 140 ● Net Income Stability



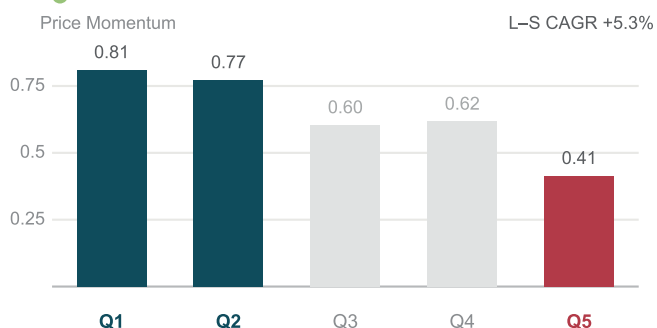
## 141 ● 1Y Chg in Free Cash Flow to Price



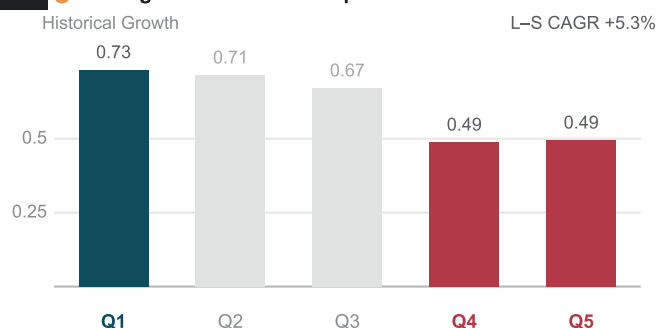
## 142 ● Net Cash Flow to Enterprise Value

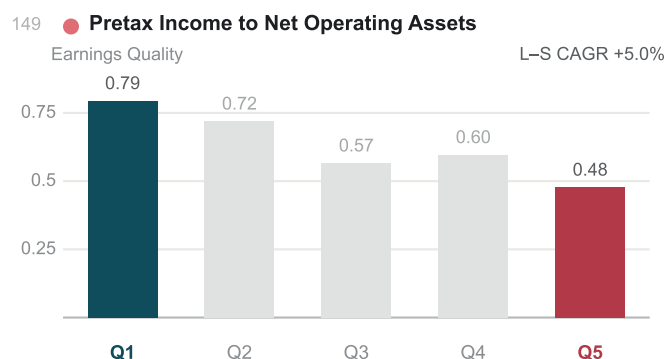
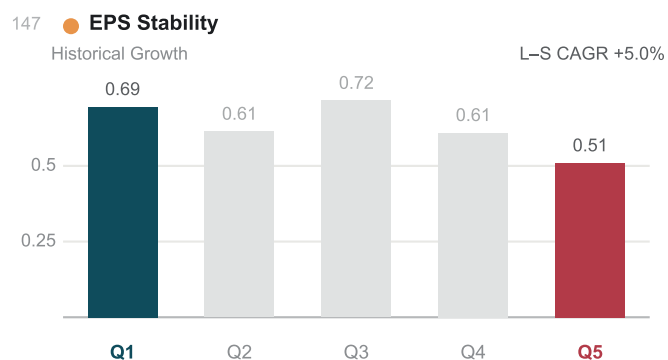
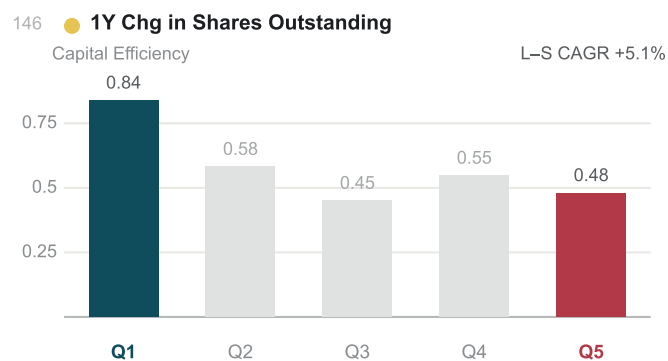
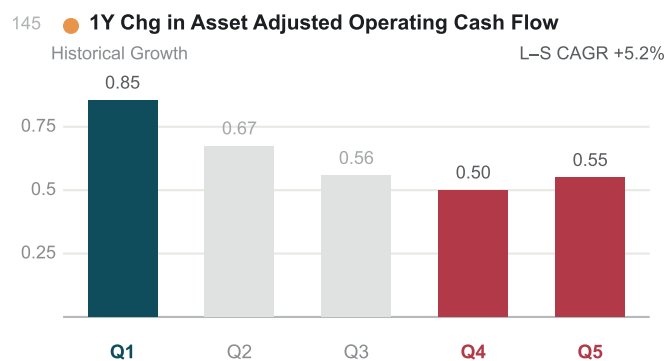


## 143 ● 12M - 1M Price Momentum

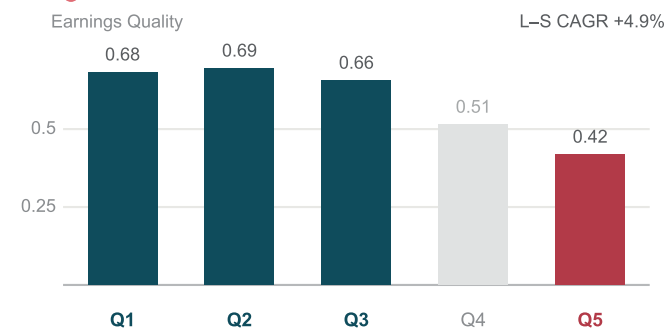


## 144 ● 1Y Chg in Free Cash Flow per Share

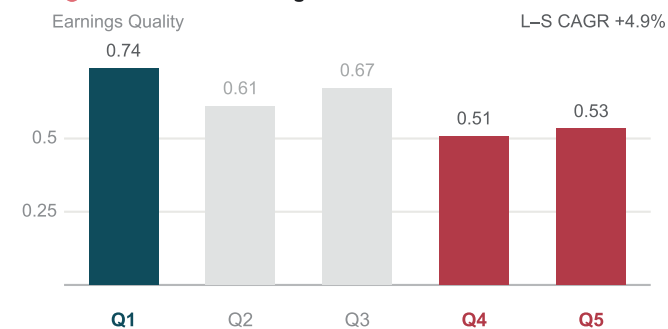




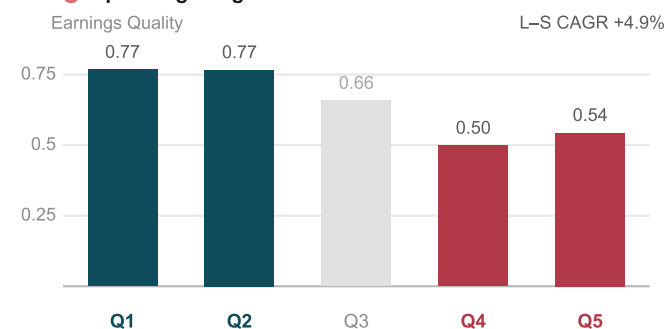
## 151 ● Free Cash Flow to Sales



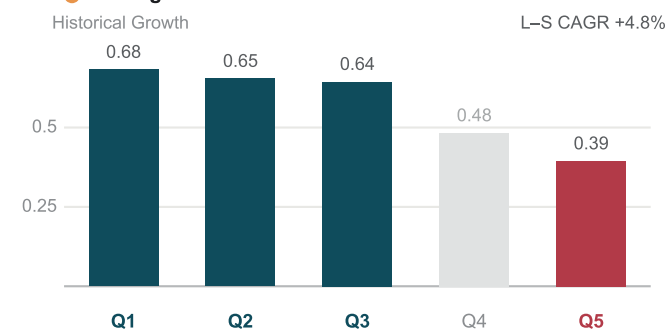
## 152 ● Year over Year Change of EPS to Sales



## 153 ● Operating Margin



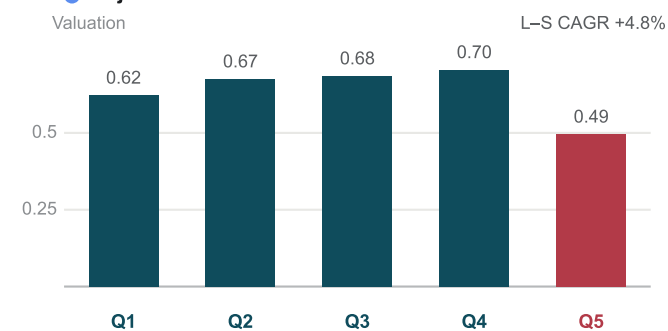
## 154 ● 3Y Chg in Sales to Price



## 155 ● EBIT Margin



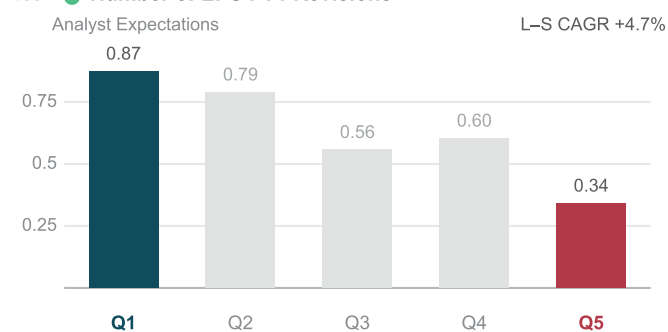
## 156 ● Adj Forward Free Cash Flow to Price Ratio



## 157 ● 3Y Chg in Free Cash Flow per Share



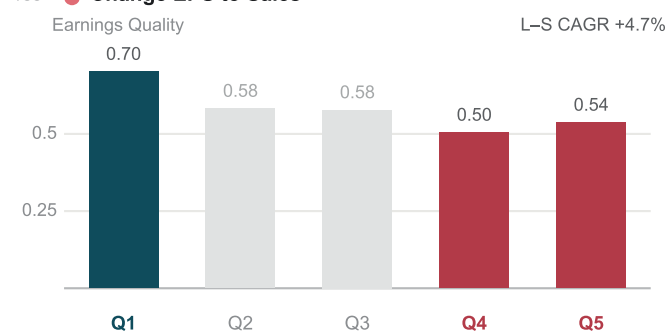
## 158 ● Number of EPS FY1 Revisions



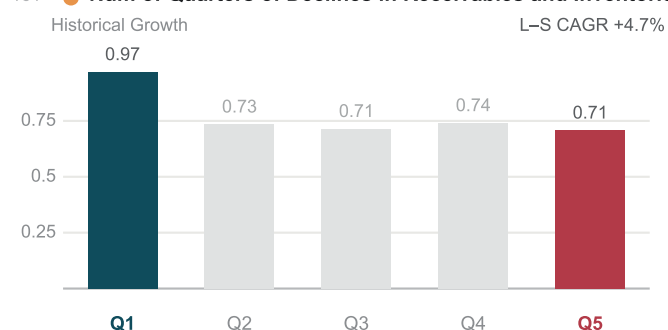
## 159 ● Working Capital to Assets Ratio



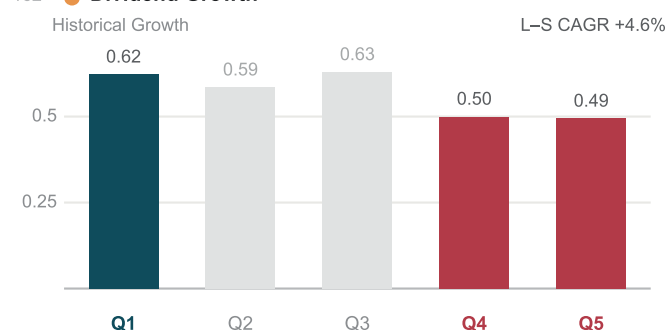
## 160 ● Change EPS to Sales

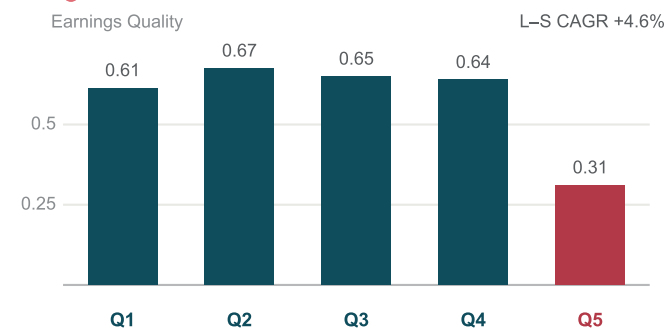
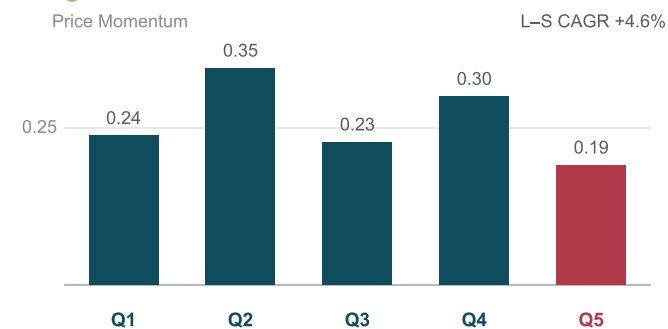
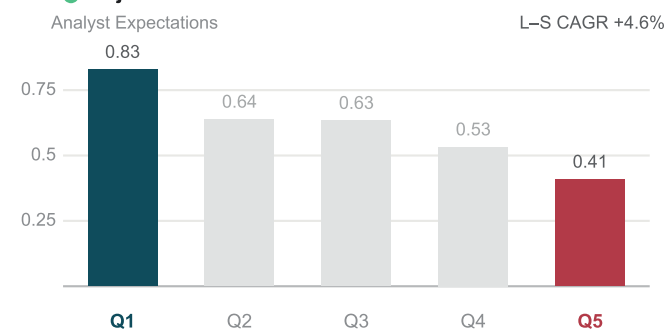
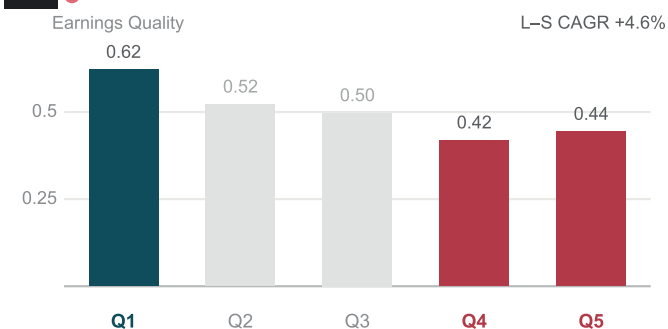
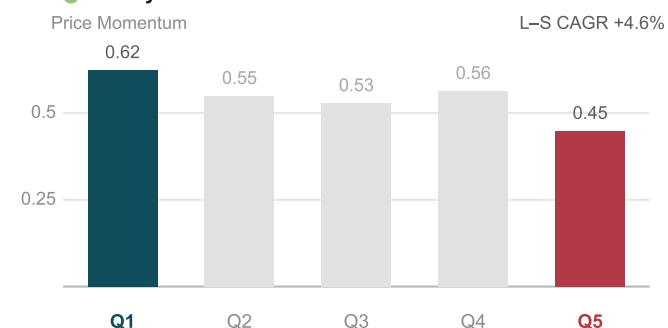
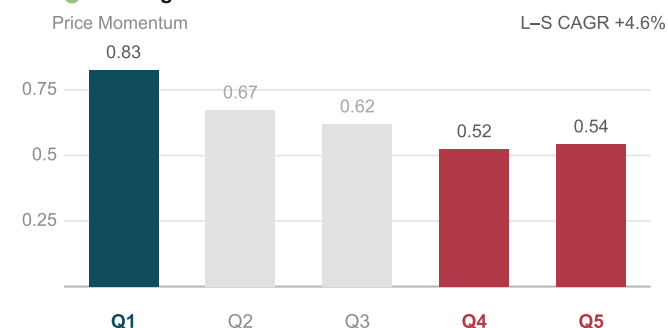


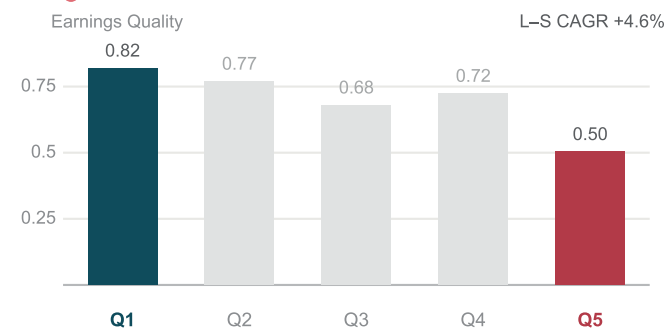
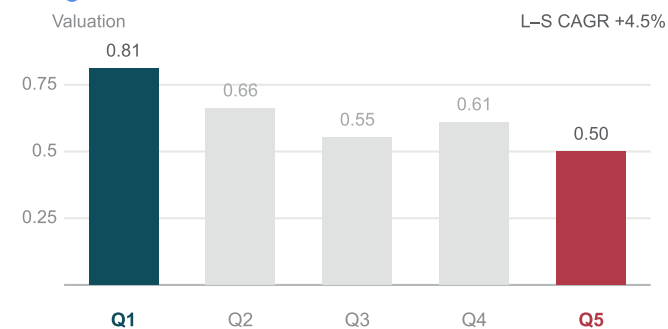
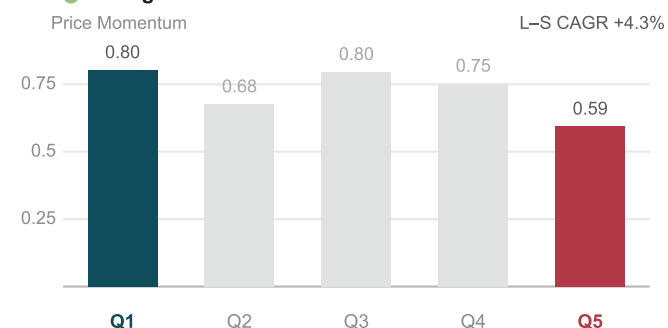
## 161 ● Num of Quarters of Declines in Receivables and Inventories/...

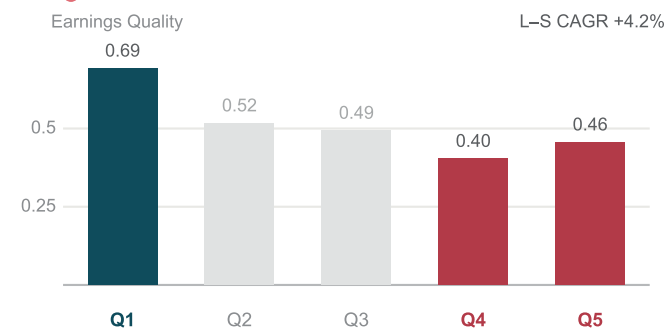
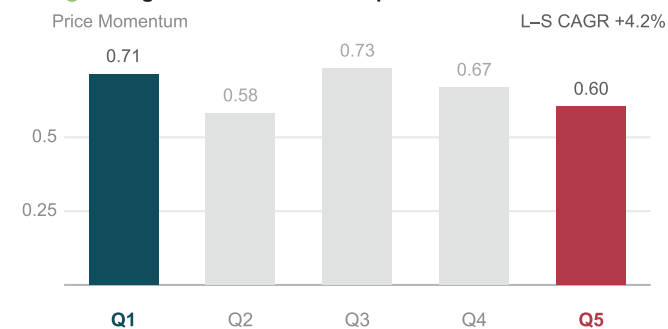
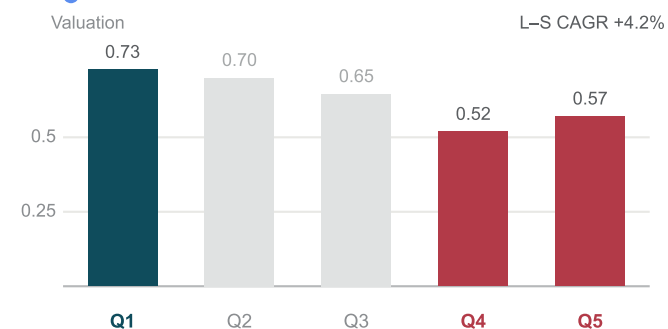
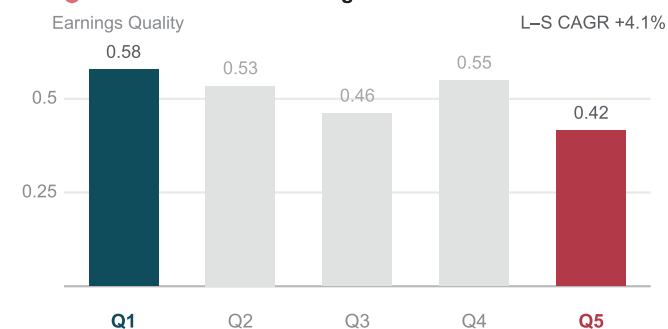
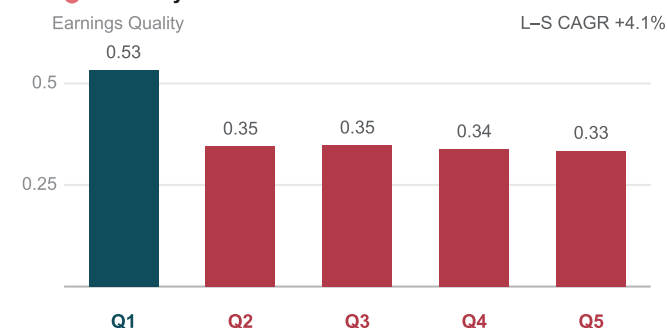


## 162 ● Dividend Growth

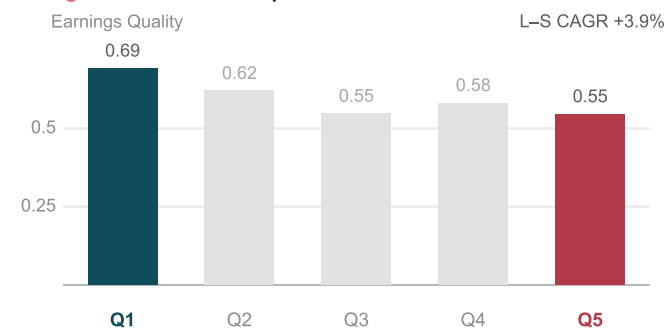


163 ● **Asset Turnover**164 ● **Share Turnover**165 ● **Adj Number of EPS FY2 Revisions**166 ● **Accrual Ratio - Cash Flows**167 ● **60 Day Coefficient of Variation of Volume to Price**168 ● **52W High Low**

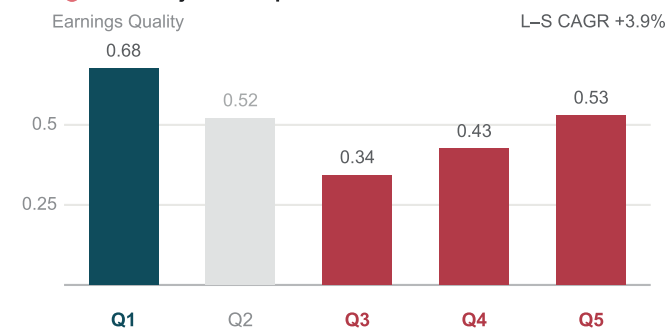
169 ● **Current Ratio**170 ● **Cash to Total Assets**171 ● **3Y Chg in Sales Turnover**172 ● **6M Momentum in Trailing 12M Sales**173 ● **Change in Number of Holders**174 ● **3Y Chg in Asset Adjusted Free Cash Flow**

175 ● **SG&A to Sales**176 ● **Foreign Institution Ownership**177 ● **Net Current Assets to Price Ratio**178 ● **Sales to Gross Profit Margin Growth**179 ● **Risk Adjusted Relative Strength**180 ● **Inventory to Assets Ratio**

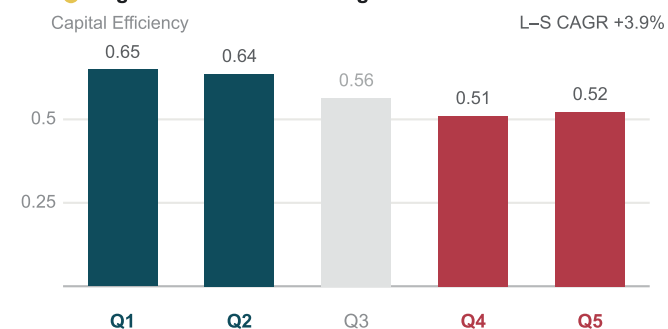
## 181 ● ROA 60 Month Slope



## 182 ● Asset Adjusted Capital Investments



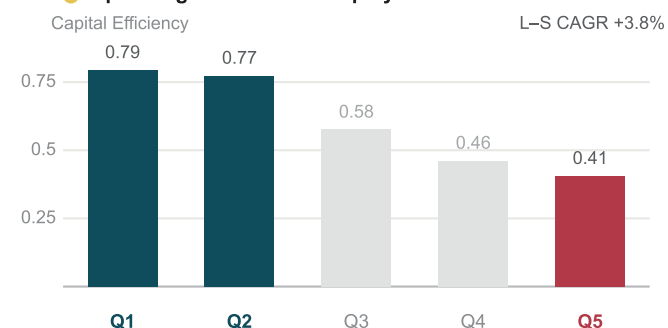
## 183 ● Degree of Financial Leverage



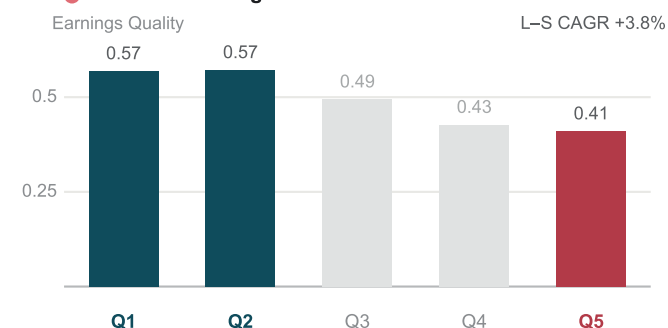
## 184 ● Return on Equity Stability



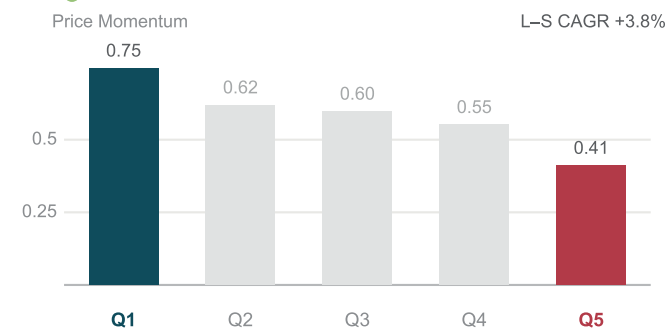
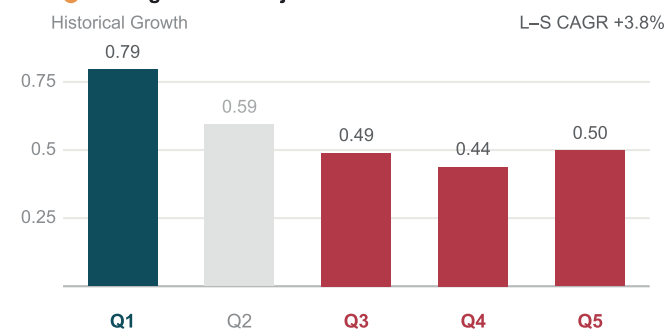
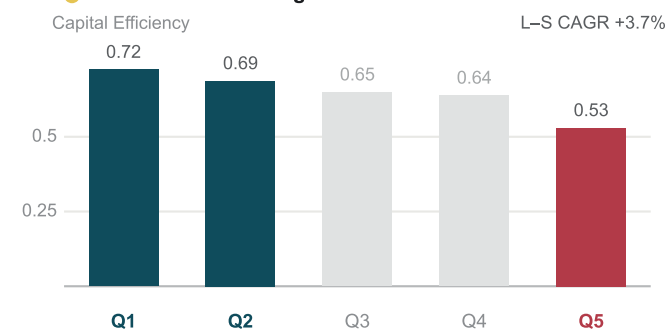
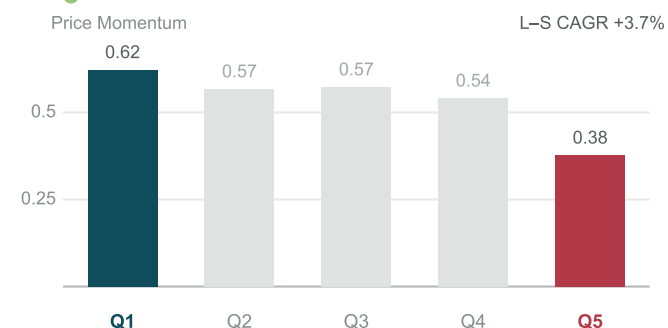
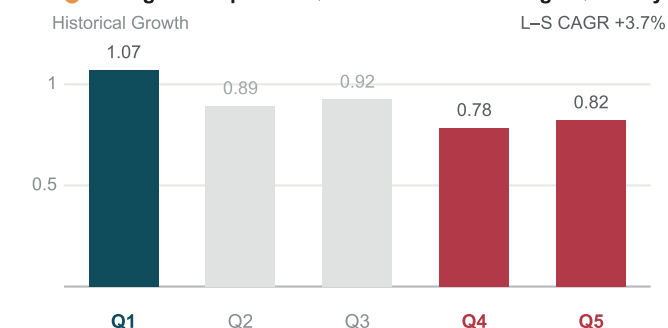
## 185 ● Operating Cash Flow to Equity

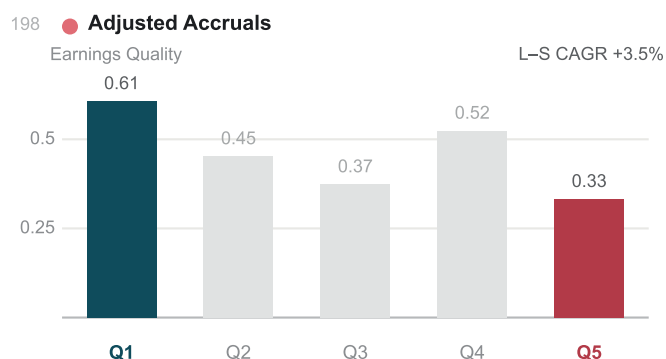
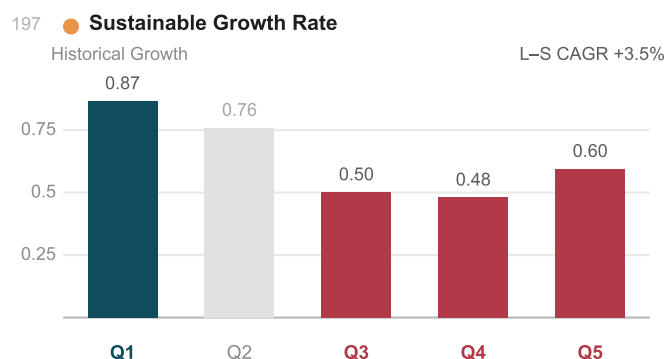
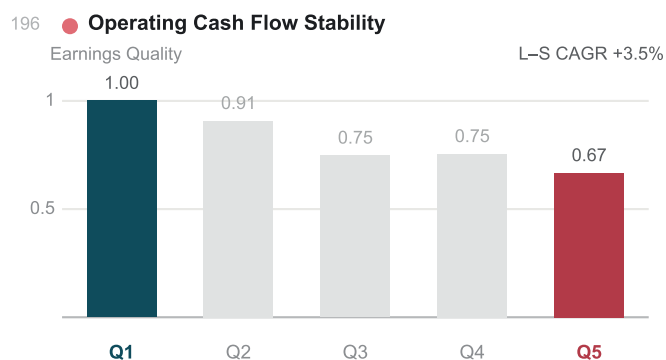
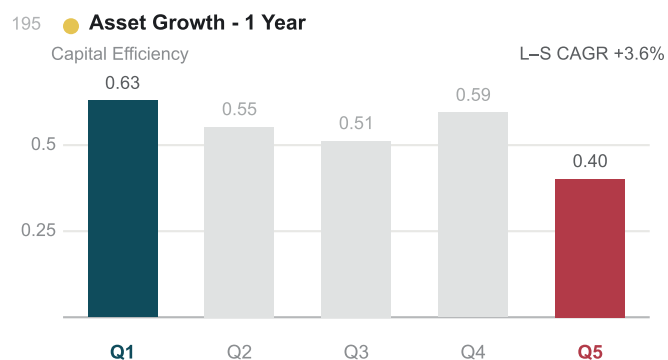
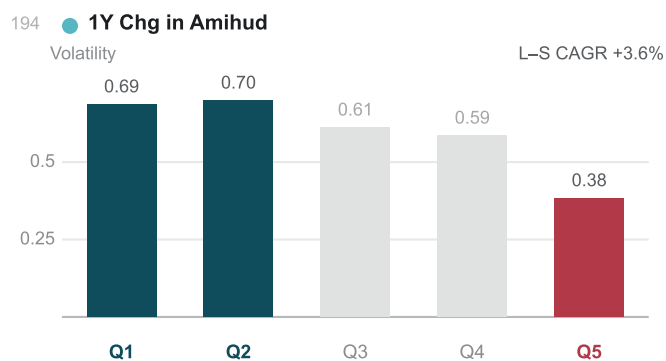
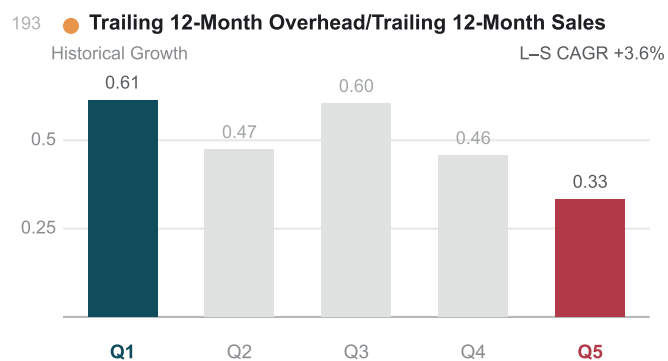


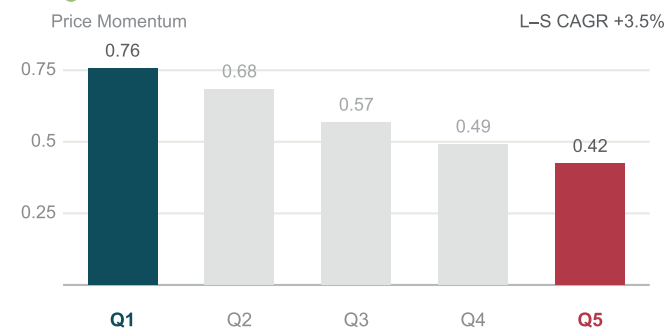
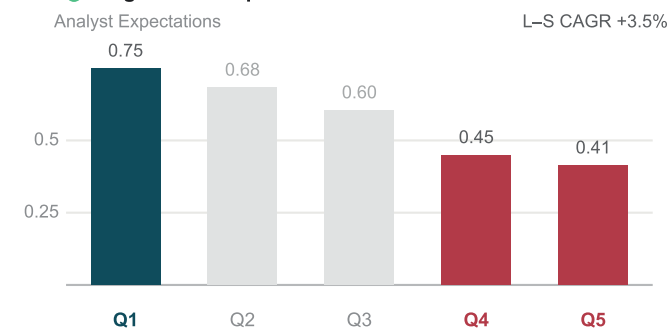
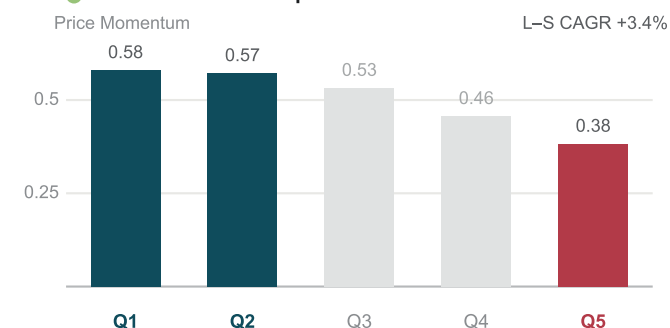
## 186 ● Retained Earnings to Total Assets





187 ● **Net Profit Margin**188 ● **26W Price Stat**189 ● **3Y Chg in Asset Adjusted EPS**190 ● **Year over Year Change of Total Debt**191 ● **24M Residual Return Variance**192 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**



199 ● **180D Price TStat**200 ● **Target Price Gap to 6M EMA**201 ● **Free Cash Flow to Common Equity**202 ● **1Y Chg in Earnings per Share**203 ● **Working Capital to Total Assets**204 ● **Institution Ownership Concentration**

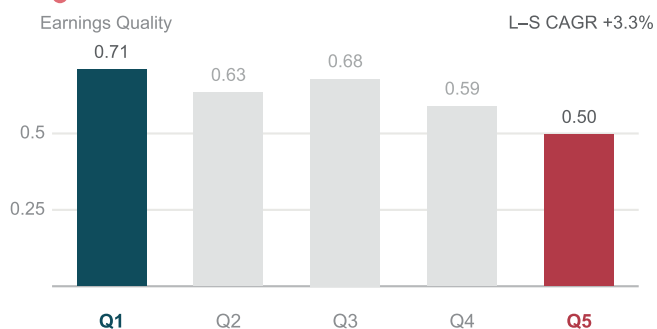
## 205 ● Return on Assets Stability



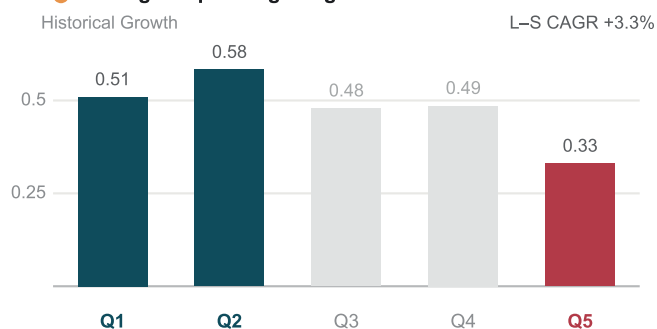
## 206 ● 3Y Chg in Earnings per Share



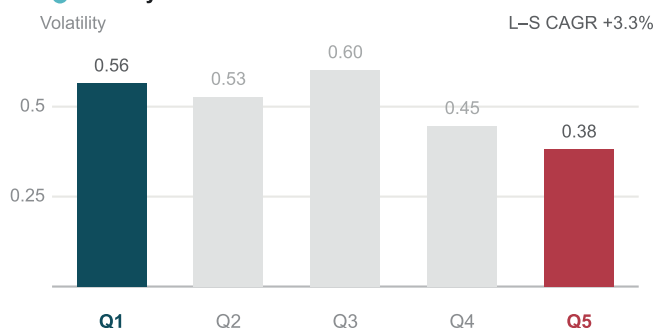
## 207 ● Accrual Ratio - Balance Sheet



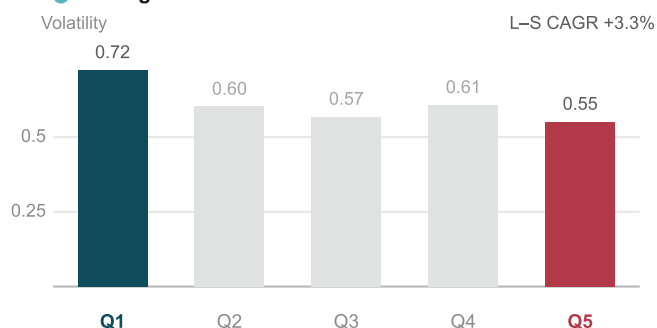
## 208 ● 3Y Chg in Operating Margin



## 209 ● 130 Day Minimum Return



## 210 ● 1Y Chg in Debt to Assets



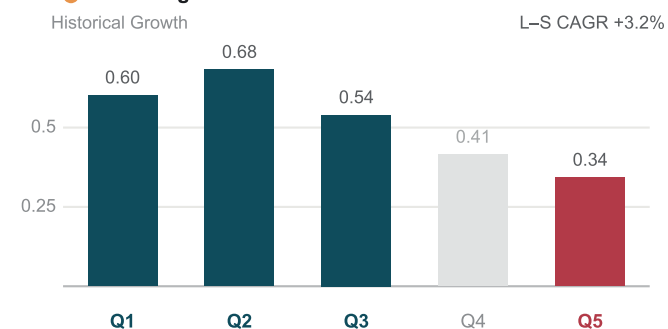
## 211 ● Ownership Breadth Stability of Institution Ownership by Buyers



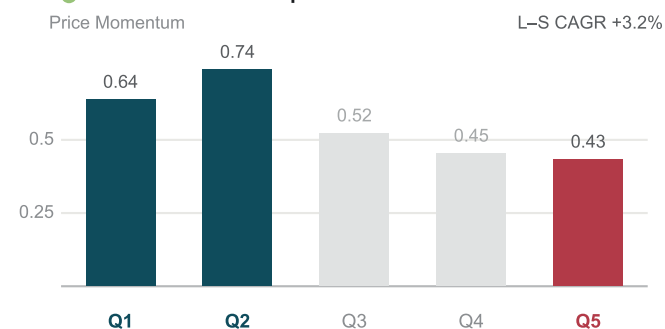
## 212 ● Pretax Profit Margin



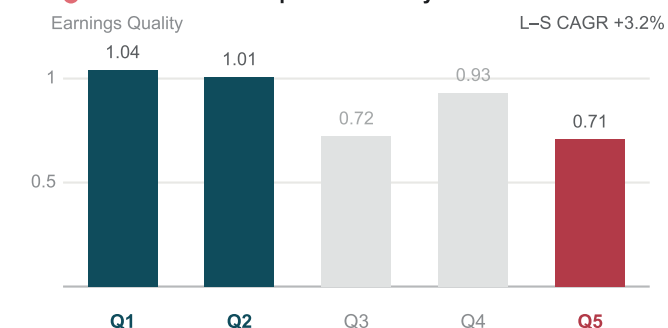
## 213 ● 3Y Average Annual Sales Growth



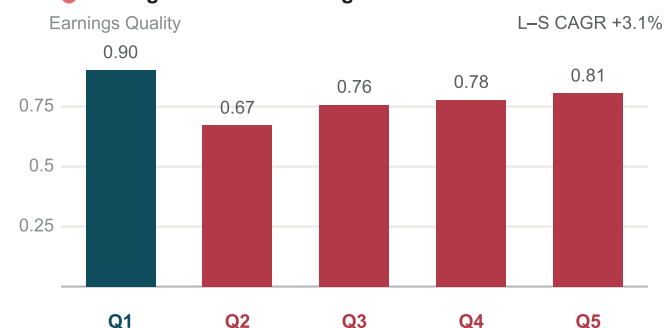
## 214 ● Institution Ownership Level

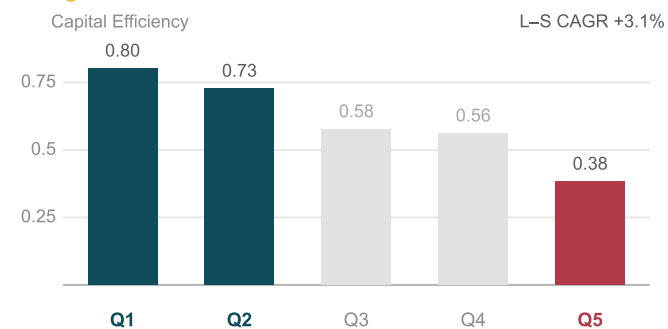
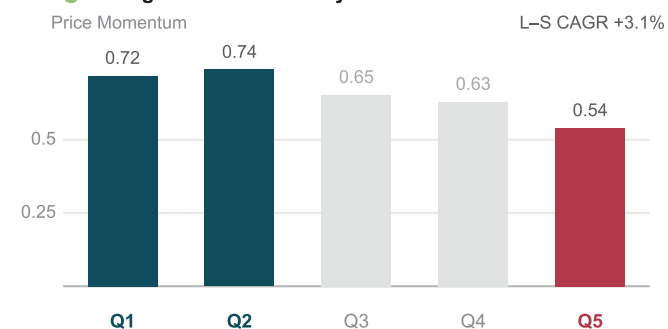
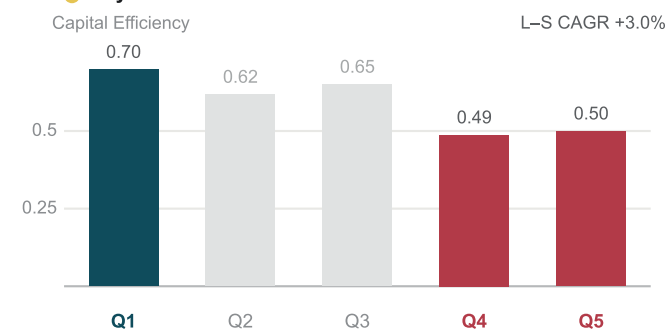
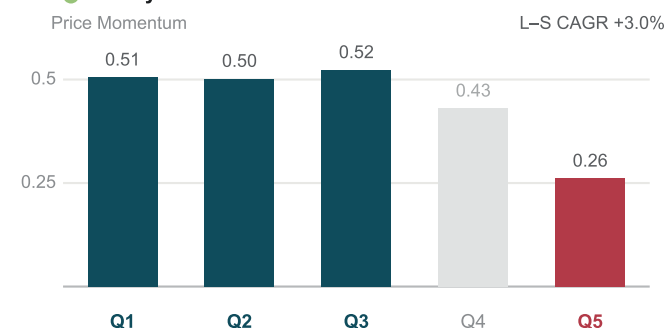


## 215 ● Research &amp; Development Intensity

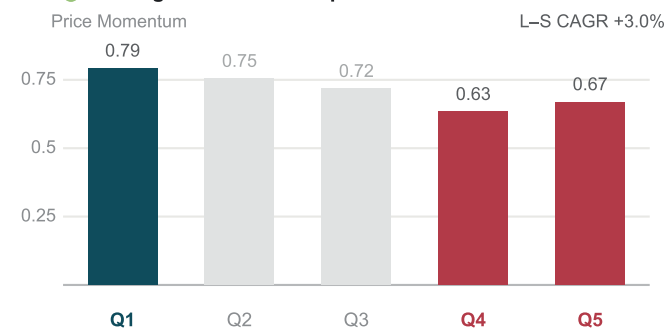


## 216 ● 1Y Chg in Sales to Earnings

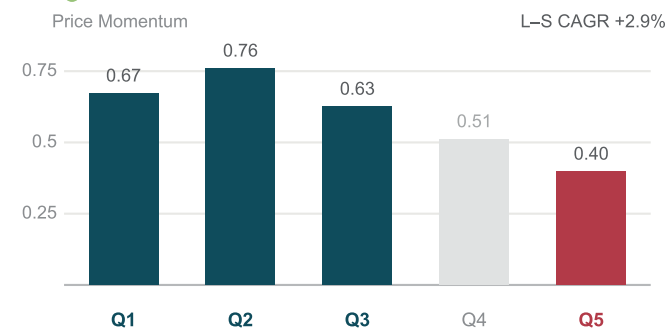


217 ● **Gross Profit to Assets**218 ● **Cash Conversion Cycle**219 ● **Change in Number of Buyers**220 ● **Payout Ratio**221 ● **10 Day MACD Trend**222 ● **Change of Sales to Change of EPS**

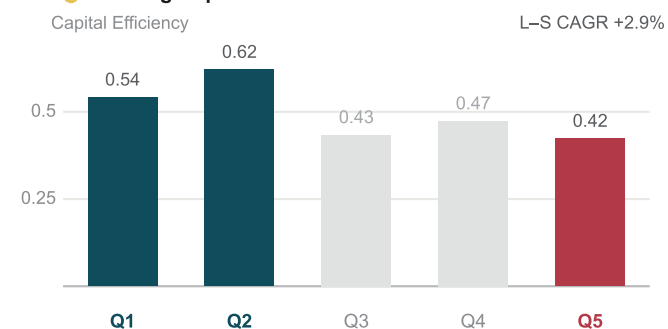
## 223 ● 6M Chg in 18M CAPM Alpha



## 224 ● Investment Duration



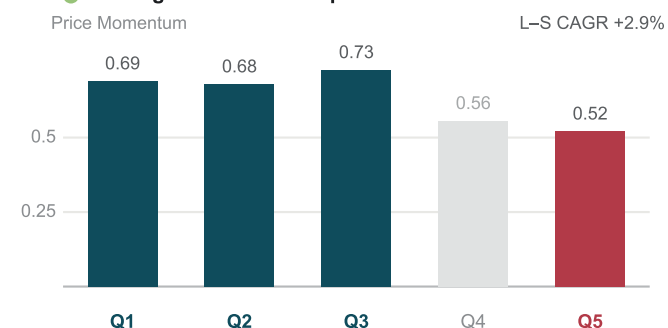
## 225 ● Working Capital Turnover Ratio



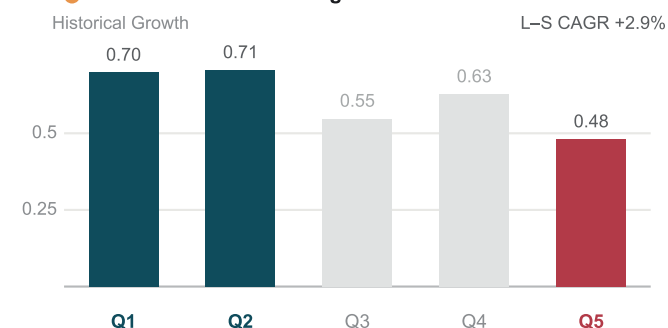
## 226 ● Ownership Breadth Stability of Institution Ownership by Hol...



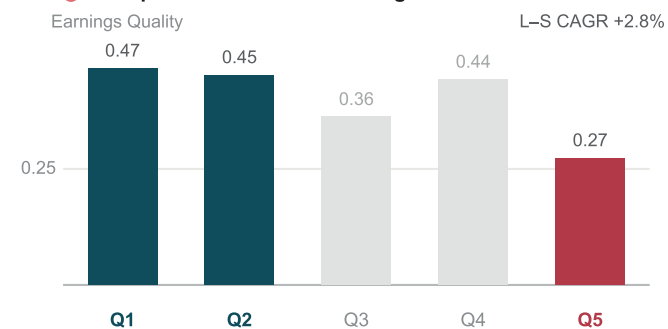
## 227 ● 6M Chg in 36M CAPM Alpha



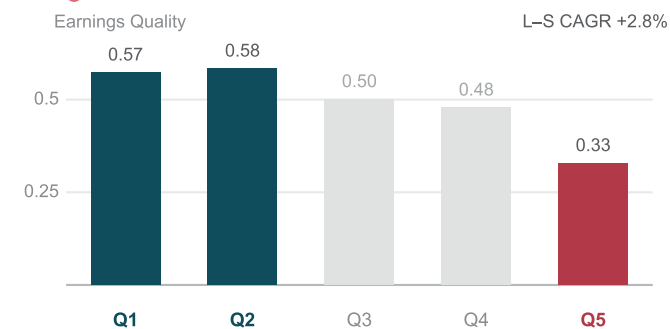
## 228 ● 3M Momentum in Trailing 12M Sales



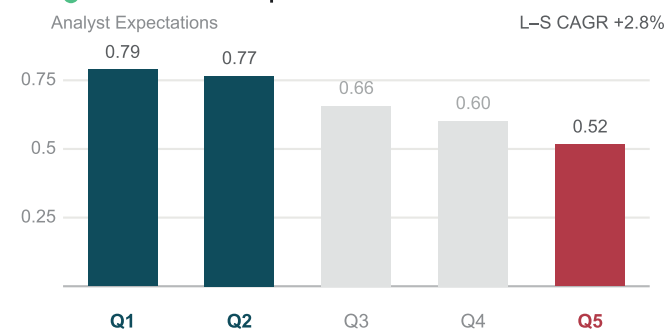
## 229 ● Unexpected Receivables Change



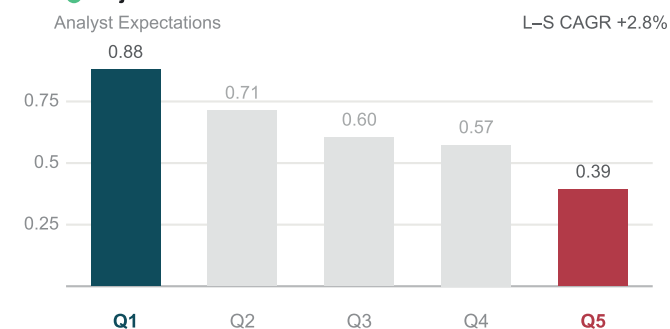
## 230 ● Cash to Sales



## 231 ● EBIT Estimate Dispersion



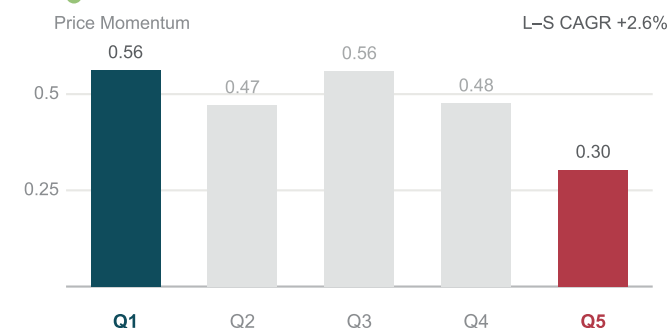
## 232 ● Adjusted Number of EPS FY1 Revisions



## 233 ● Volatility adjusted 12M return

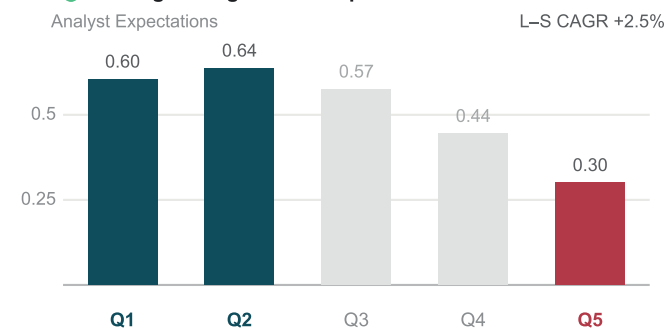


## 234 ● Standardized 36M Residual





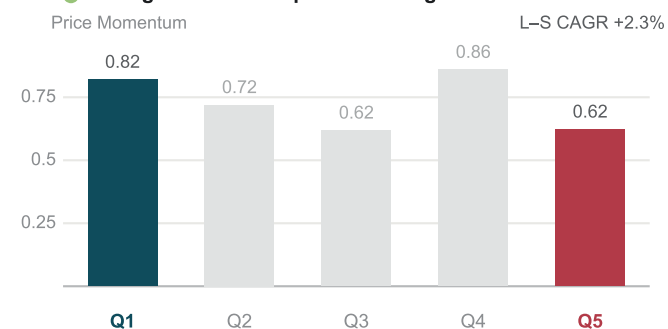
## 235 ● 6M Chg in Target Price Gap



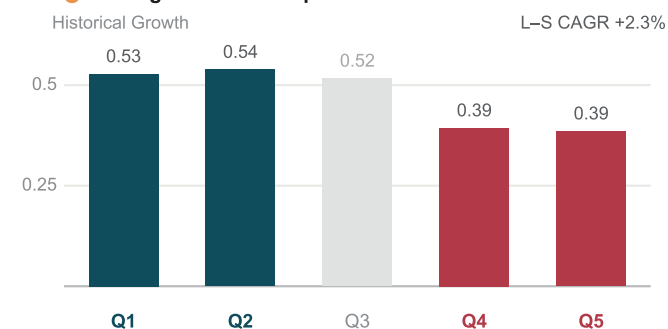
## 236 ● 1Y Chg in EPS to Operating Cash Flow



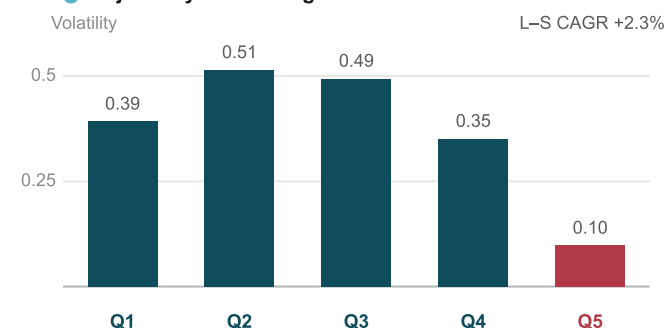
## 237 ● Change of Ownership Level - Hedge Fund



## 238 ● 1Y Chg in Cash Flow per Share

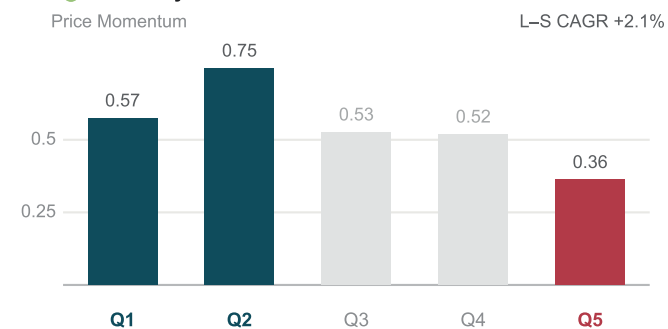
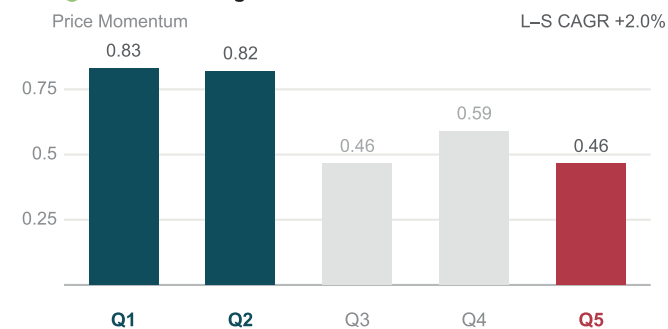
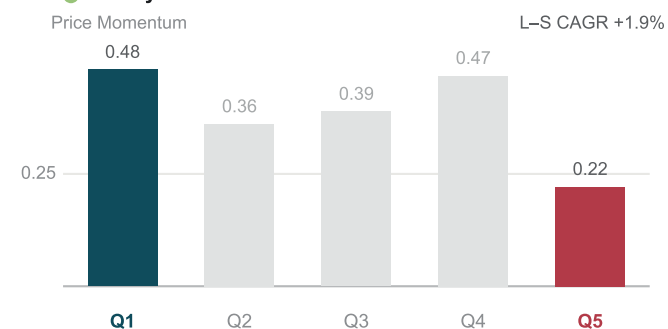
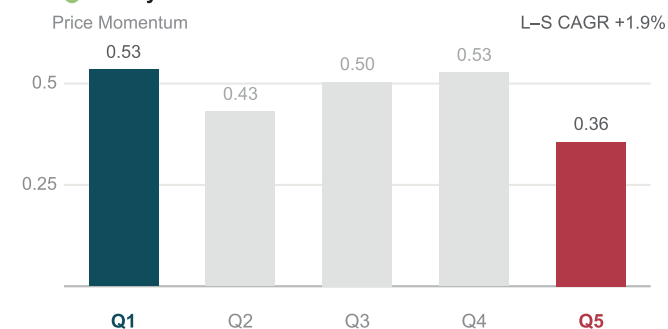
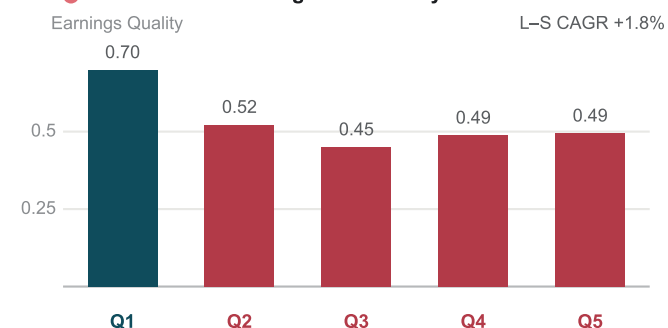
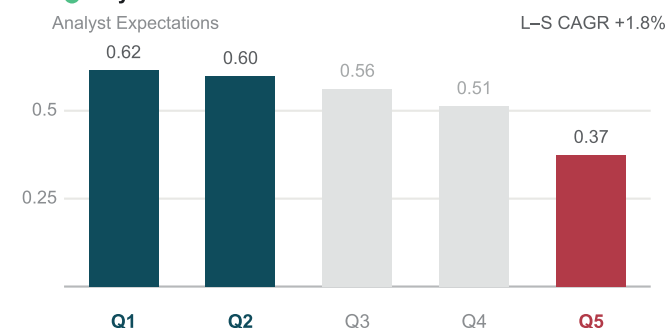


## 239 ● Adj 50 Day Volume Signal



## 240 ● Analyst Earnings Estimate Diffusion

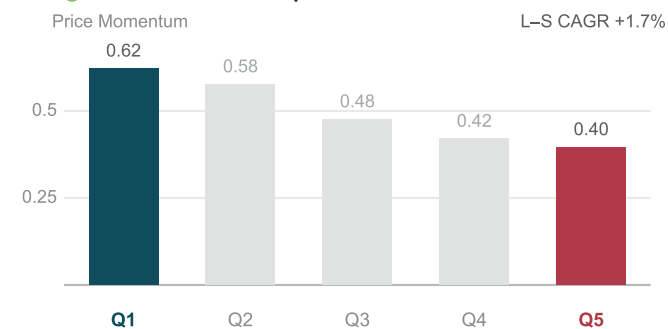


241 ● **Max Daily Return in the Past 6 Months**242 ● **6 Month Average Share Turn Over**243 ● **30 Day Coefficient of Variation of Volume to Price**244 ● **20 Day Coefficient of Variation of Volume to Price**245 ● **Year over Year Change of Inventory to Assets**246 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

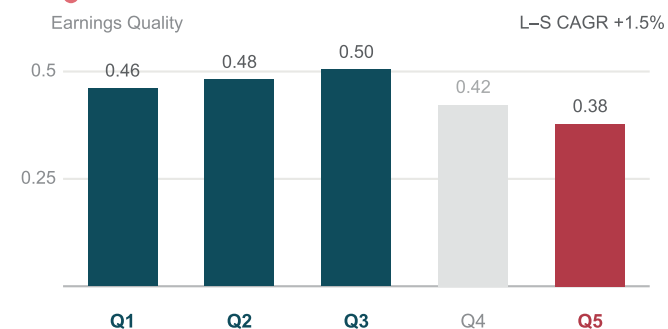
## 247 ● Closing Price to 52 Week High



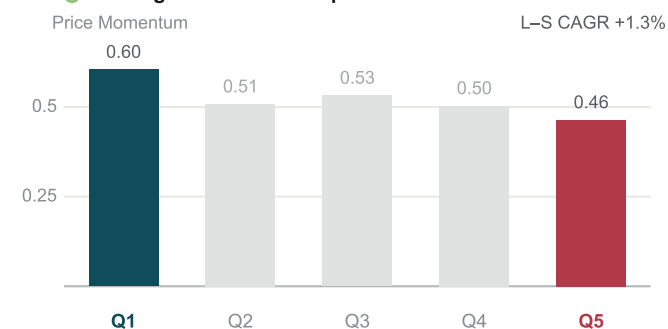
## 248 ● Institution Ownership Turnover



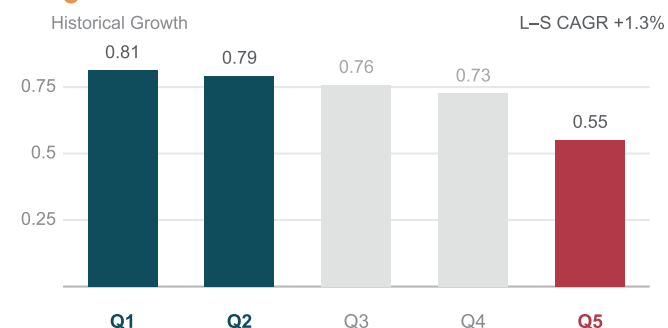
## 249 ● ROE 20 Qtr Standard Deviation



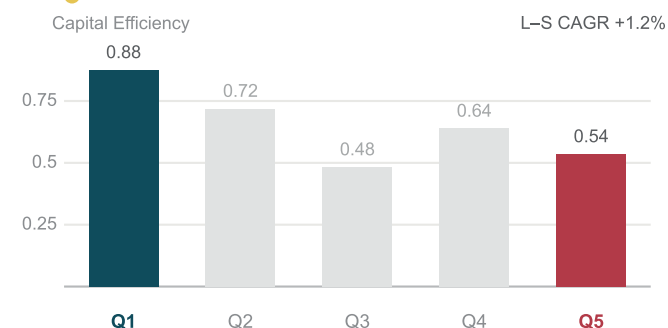
## 250 ● 6M Chg in 12M CAPM Alpha



## 251 ● Sales Acceleration



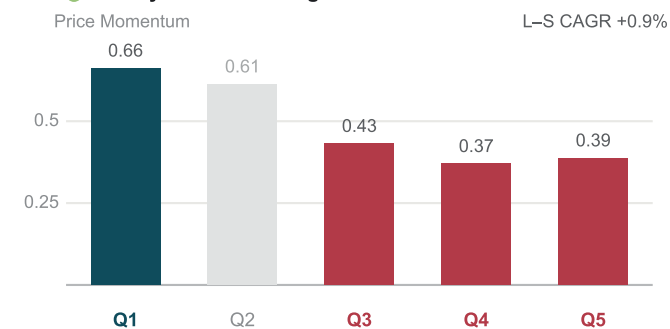
## 252 ● Normalized ROA



## 253 ● 6M Avg Chg 1M Recommendation



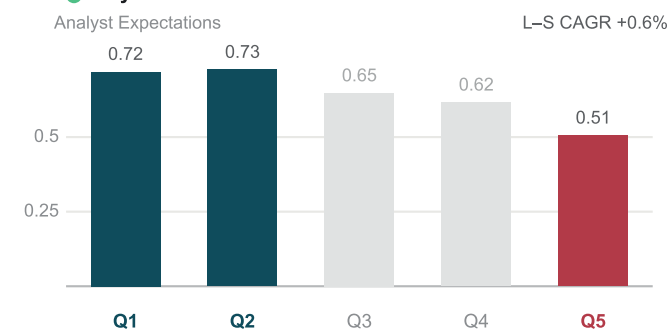
## 254 ● 14-Day Relative Strength Index



## 255 ● Change in TTM Depr. to Capex



## 256 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

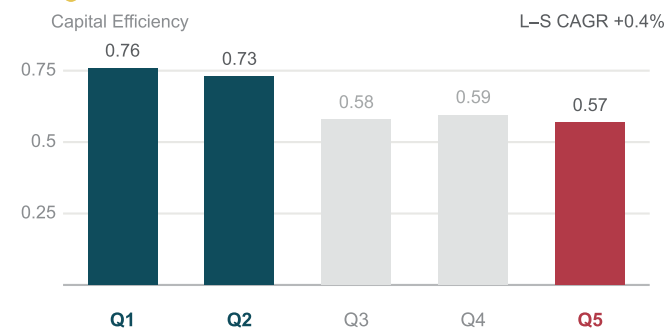
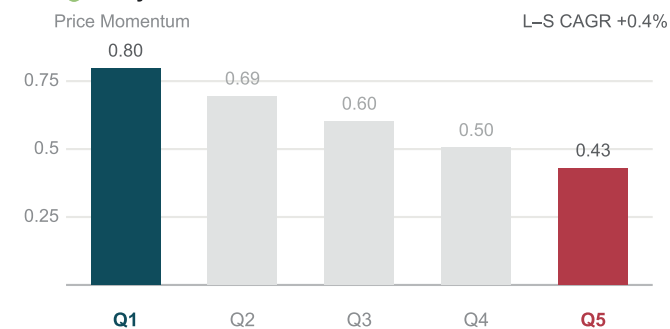
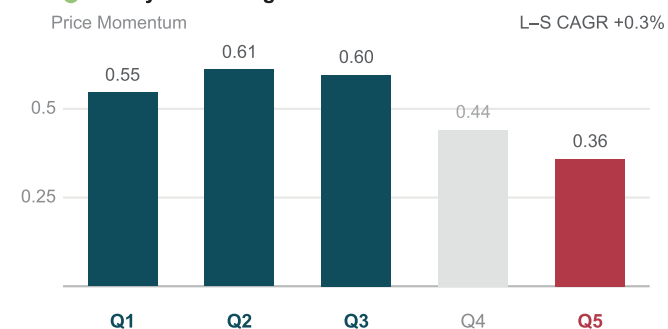
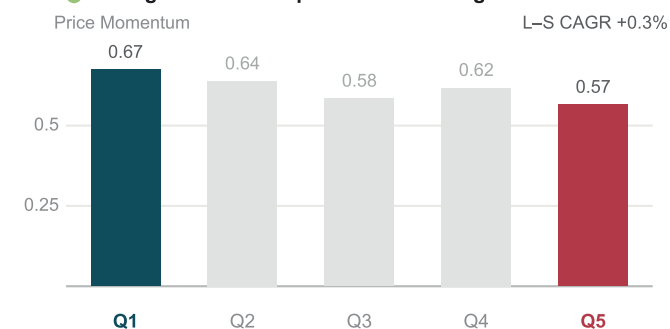
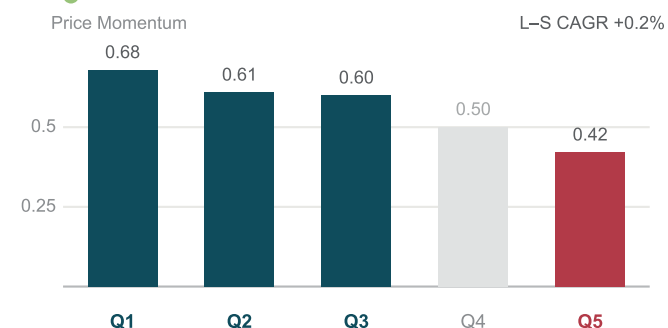


## 257 ● Revenue Estimate Dispersion



## 258 ● 8-Week Change in 12-Month Forward Earnings Consensus Estima...



259 ● **Normalized ROE**260 ● **5 Day Price Reversal**261 ● **50 Day Volume Signal**262 ● **Change of Ownership Level - All Managers**263 ● **1M Price Reversal**264 ● **1Y Chg in Share Turnover**